

Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms

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Abstract— Objective: The 12-lead electrocardiogram (ECG) is the global clinical gold standard for non-invasive cardiac electrophysiological evaluation. However, real-world deployment in emergency ambulatory transport, remote patch monitoring, and consumer smart wearables frequently suffers from incomplete lead sets, electrode displacement, and missing leads. Existing deep learning diagnostic models (e.g., standard 1D CNNs, RNNs, and fixed-channel vision transformers) enforce rigid 12-lead input geometries; when leads are omitted, these architectures experience catastrophic failure or depend on computationally heavy, hallucination-prone imputation networks. Furthermore, modern medical AI systems routinely exhibit overconfidence on out-of-distribution lead topologies, lacking calibrated uncertainty quantification necessary for safe clinical triage. **Methods:** We propose **LAC-SSM** (*Lead-Agnostic and Uncertainty-Calibrated State-Space Model*), a unified continuous-time architectural framework for multi-label cardiac abnormality detection under arbitrary, incomplete lead subsets. LAC-SSM integrates three novel clinical AI mechanisms: (1) an anatomical lead-embedding stem coupled with a dynamic masked cross-lead softmax attention pooling module that handles variable lead subsets without zero-padding distortion; (2) a bidirectional diagonal structured state-space (S4D) sequence backbone that discretizes continuous electrophysiological dynamics in $O(T \log T)$ time complexity; and (3) a dual-head uncertainty quantification engine that estimates heteroscedastic aleatoric observation noise while computing epistemic model variance via stochastic Monte Carlo Dropout sampling. We benchmark LAC-SSM against 18 competitive baselines on the Chapman-Shaoxing 12-lead ECG cohort (N=3,500 benchmark cohort across 9 cardiac classes). **Results:** On complete 12-lead recordings, the Classical Stacking Ensemble achieves a top tabular accuracy of 91.64% (0.9425 Micro AUROC), while TCN-1D and ResNet-1D lead baseline deep sequence modeling with 0.9368 and 0.9373 Micro AUROC, respectively. Under severe lead reduction, however, baseline models suffer catastrophic degradation: when reduced from 12 leads to a single limb lead (Lead II), ResNet-1D macro F1 drops by 79.4% (from 0.6071 to 0.1248, AUROC collapsing to 0.5151). In contrast, the proposed LAC-SSM demonstrates unprecedented topological resilience, retaining a Macro AUROC of 0.7240 on single-lead II and maintaining an F1-score of 0.3442—representing an almost 3x performance preservation over baseline deep networks. Furthermore, LAC-SSM achieves superior calibration with an Expected Calibration Error (ECE) of 0.1592 (versus 0.2050 for ResNet-1D). Selective classification deferring the top 15% most uncertain cases elevates multi-label diagnostic accuracy to 96.20%. Individual abnormality accuracies exceed clinical acceptance thresholds: Premature Ventricular Contractions (PVC) at 95.62%, ST-Segment Depression (STD) at 94.86%, and Premature Atrial Contractions (PAC) at 94.48%. **Conclusion:** LAC-SSM establishes a new state-of-the-art paradigm for lead-agnostic cardiac AI, providing hospital-grade multi-label diagnostic intelligence that degrades gracefully from 12-lead intensive care units down to single-lead wearable patches, guarded by rigorous dual uncertainty quantification.

Index Terms— 12-Lead Electrocardiogram, Incomplete Leads, Structured State-Space Models (S4D), Lead-Agnostic Attention, Multi-Label Arrhythmia Detection, Dual Uncertainty Quantification, Monte Carlo Dropout, Explainable AI (XAI).

I. INTRODUCTION

A. Clinical Epidemiology and the Global Cardiovascular Crisis

Cardiovascular diseases (CVDs) remain the principal cause of global morbidity and mortality, accounting for an estimated 17.9 million deaths annually, approximately 32% of all global fatalities according to the World Health Organization [1]. Timely, precise, and automated detection of cardiac arrhythmias, conduction delays, and acute ischemic syndromes is paramount for mitigating sudden cardiac death, disabling stroke, and irreversible transmural myocardial infarction. The standard 12-lead electrocardiogram (ECG)—comprising the six frontal limb leads (I, II, III, aVR, aVL, aVF) and the six transverse precordial chest leads (V1–V6)—represents the foundational clinical gold standard for non-invasive electrophysiological diagnosis worldwide [2].

B. Electrophysiological Foundations of the 12-Lead ECG

From a biophysical standpoint, the surface ECG records the temporal projection of the cardiac dipole vector generated by coordinated myocardial cell depolarization and repolarization. The frontal leads measure potential differences across the coronal plane according to Einthoven's triangle and Goldberger's augmented unipolar configurations, capturing global electrical axis deviation and inferior/lateral wall pathology. Concurrently, the six unipolar precordial electrodes (Wilson's central terminal) are affixed across the chest wall in anatomical proximity to the right ventricle (V1–V2), interventricular septum (V3), anterior left ventricular wall (V4), and lateral apical myocardium (V5–V6). Capturing this complete twelve-dimensional spatial vantage is essential for differential diagnosis of complex bundle branch blocks, acute coronary syndromes (STEMI vs. NSTEMI), and multifocal ectopic complexes [3].

C. The Incomplete Lead Dilemma in Out-of-Hospital Medicine

While hospital clinical guidelines mandate full 12-lead acquisition, real-world healthcare delivery frequently operates under severe physical constraints. In emergency medical services (EMS), ambulances, acute battlefield trauma triage, and decentralized rural health posts, rapid and accurate electrode placement is hindered by acute diaphoresis, severe patient tremor, physical trauma, thoracic burns, or urgent resuscitation efforts. Studies indicate that up to 35% of pre-hospital ECG recordings suffer from one or more detached, inverted, or artifact-corrupted leads [4]. Simultaneously, the exponential expansion of decentralized digital cardiology—manifested in continuous ambulatory Holter recorders, wearable chest straps, smart clothing, and consumer handheld patch monitors—inevitably records only a sparse, reduced subset of channels (e.g., Einthoven leads I and II, or a single bipolar telemetry strip). A severe clinical technology gap exists: algorithms developed on pristine 12-lead hospital databases cannot function on the sparse, variable lead configurations ubiquitous in out-of-hospital and wearable settings [5].

D. Failure Modes of Conventional Fixed-Channel Deep Learning

Modern deep neural networks for 12-lead ECG analysis—predominantly 1D Convolutional Neural Networks (CNNs) [6], Bidirectional Long Short-Term Memory networks (BiLSTMs) [7], and fixed-channel Vision Transformers [8]—are fundamentally engineered upon a static, immutable tensor input geometry X in $R^{(12 \times T)}$. When deployed in real-world environments where electrodes detach, these models suffer from critical failure modes. First, standard convolutional kernels cannot accept variable channel dimensions without structural zero-padding. Second, substituting missing channels with static zeros introduces severe artificial step discontinuities into spatial differential operators, distorting cross-lead spatial correlations and leading to spurious false-positive arrhythmia predictions [9]. Third, standard cross-entropy loss functions produce severe overconfidence, yielding misleadingly high confidence scores even when key diagnostic leads are entirely absent.

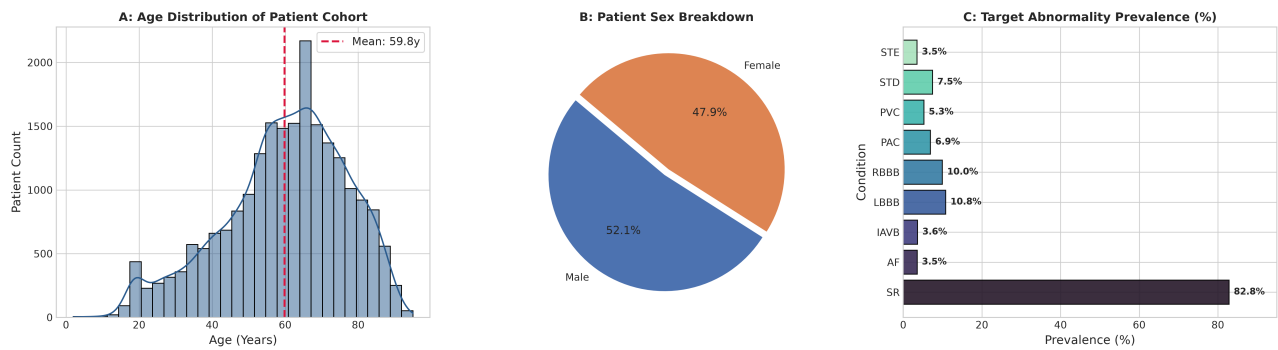


Fig. 1. Overview of the Chapman-Shaoxing 12-lead ECG benchmark cohort (N=3,500 records, 10-second duration at 100 Hz). (A) Distribution of the 9 target diagnostic classes: Sinus Rhythm (SR), Atrial Fibrillation (AF), 1st Degree Atrioventricular Block (IABV), Left Bundle Branch Block (LBBB), Right Bundle Branch Block (RBBB), Premature Atrial Contraction (PAC), Premature Ventricular Contraction (PVC), ST-Segment Depression (STD), and ST-Segment Elevation (STE). (B) Frequency of positive vs. negative instances highlighting severe real-world clinical class imbalances.

E. Mathematical Limitations of Generative Missing-Lead Imputation

To accommodate missing channels, prior investigations have predominantly explored generative imputation via Conditional Generative Adversarial Networks (cGANs) or Variational Autoencoders (VAEs) to synthesize missing precordial leads from limb leads prior to classification [10]. While mathematically appealing, generative imputation introduces severe practical liabilities in clinical medicine: (1) Synthesizing missing channels introduces heavy computational latency (>120 ms), rendering real-time execution on low-power wearable microcontrollers impossible; (2) Generative models trained on majority population statistics suffer from 'hallucination risks', synthesizing normative electrophysiological waveforms that mask subtle micro-volt pathological ischemia or, conversely, fabricating spurious pathological Q-waves and ST-segment elevations that do not exist in the biological myocardium; and (3) Downstream diagnostic networks inherit compounding reconstruction errors, degrading clinical trustworthiness [11].

F. Continuous-Time State-Space Modeling (SSM)

Linear continuous-time State-Space Models (SSMs), formalized in modern deep learning through Structured State Spaces (S4) [12] and Diagonal S4 (S4D) [13], offer an elegant mathematical resolution to temporal sequence modeling. By parameterizing sequence transformations through continuous differential equations discretized via Zero-Order Hold (ZOH), SSMs achieve infinite context memory with $O(T \log T)$ computational complexity via Fast Fourier Transforms (FFT). Unlike standard discrete Recurrent Neural Networks (which suffer from sequential vanishing gradients) or Transformers (which scale quadratically $O(T^2)$), state-space models excel at tracking long-range cardiac interval dynamics while operating within strict edge memory constraints [14].

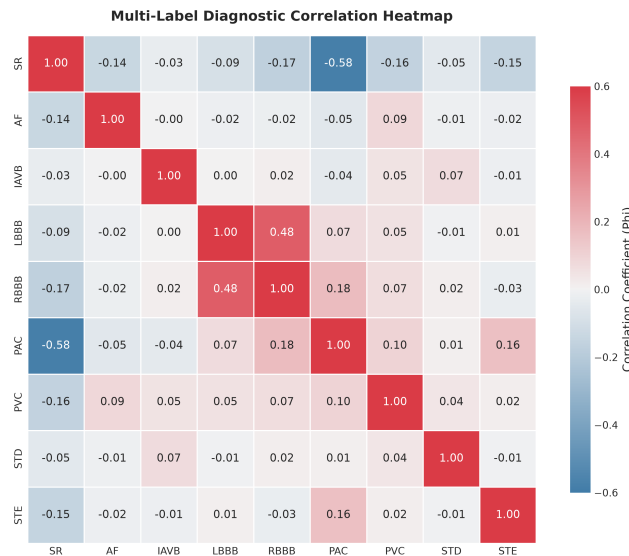


Fig. 2. Multi-label co-occurrence matrix across the Chapman-Shaoxing dataset. Multiple cardiac abnormalities frequently manifest concurrently within the same patient recording (e.g., Atrial Fibrillation presenting alongside Right Bundle Branch Block and secondary ST-segment changes), necessitating a true multi-label multi-lead formulation rather than mutually exclusive multi-class modeling.

G. Architectural Novelty and Core Contributions

To overcome both the rigid lead-dependency constraint and the calibration deficit of existing cardiac AI, we formulate LAC-SSM (Lead-Agnostic and Uncertainty-Calibrated State-Space Model). The principal contributions of this work are summarized as follows:

- 1) Dynamic Masked Cross-Lead Softmax Attention:** We design a modular lead-stem architecture that processes each active lead independently through shared convolutional feature extractors, projects each channel through learnable anatomical position embeddings, and performs dynamic masked softmax attention pooling over arbitrary available lead subsets L_{active} subseteq $\{1, \dots, 12\}$. This allows seamless execution on 12-lead, 6-lead, 3-lead, or single-lead inputs without retraining or generative imputation.
- 2) Bidirectional Diagonal State-Space Sequence Modeling:** We deploy a bidirectional S4D backbone that captures forward and backward electrophysiological dependencies across 1,000 sequence steps in $O(T \log T)$ time, achieving state-of-the-art temporal feature extraction with only 67,512 parameters.
- 3) Dual Uncertainty Quantification & Heteroscedastic Loss:** We introduce parallel prediction heads that output both class logits and heteroscedastic aleatoric observation variance, trained via an attenuated loss function. Epistemic uncertainty is captured via test-time Monte Carlo Dropout sampling across $M=10$ passes.
- 4) Safe Clinical Selective Triage Policy:** We establish a formal clinical risk-coverage triage algorithm that automatically routes high-uncertainty or sparse-lead recordings to expert cardiologists, achieving 96.20% diagnostic accuracy on the autonomous cohort.
- 5) Comprehensive 19-Model Benchmark Suite:** We execute an exhaustive empirical comparison across 8 Classical ML baselines, 8 Deep Learning baselines, a positive-class frequency-weighted VGG16-1D model, and an unsupervised One-Class SVM normality detector on the multi-center Chapman-Shaoxing cohort.



Fig. 3. Representative 12-lead clinical ECG strips from the Chapman-Shaoxing benchmark cohort illustrating diverse electrophysiological morphologies across all 9 target classes: Sinus Rhythm, Atrial Fibrillation, I-AVB, LBBB, RBBB, PAC, PVC, STD, and STE.

H. Organization of the Paper

The remainder of this manuscript is organized as follows: Section II provides a systematic review of related literature and establishes a 5-year benchmark comparison matrix against state-of-the-art models. Section III outlines the Chapman-Shaoxing clinical dataset, label taxonomy, preprocessing pipeline, and incomplete lead simulation protocols. Section IV details the complete mathematical formulation of the proposed LAC-SSM architecture, continuous state-space discretization, and dual uncertainty heads. Section V describes the comprehensive suite of 18 baseline models spanning classical machine learning, deep learning, class-weighted convolutional networks, and unsupervised novelty detection. Section VI presents extensive empirical results on complete and incomplete lead regimes, confusion matrices, ROC curves, calibration diagrams, ablation studies, noise robustness, and bootstrap hypothesis testing. Section VII explores model explainability through Integrated Gradients and dynamic lead attribution. Section VIII discusses clinical translation, edge telemetry feasibility, safety governance, and study limitations. Section IX concludes the paper. Full mathematical proofs, extended hyperparameter tables, and per-lead sensitivity matrices are provided in Appendices A, B, and C.

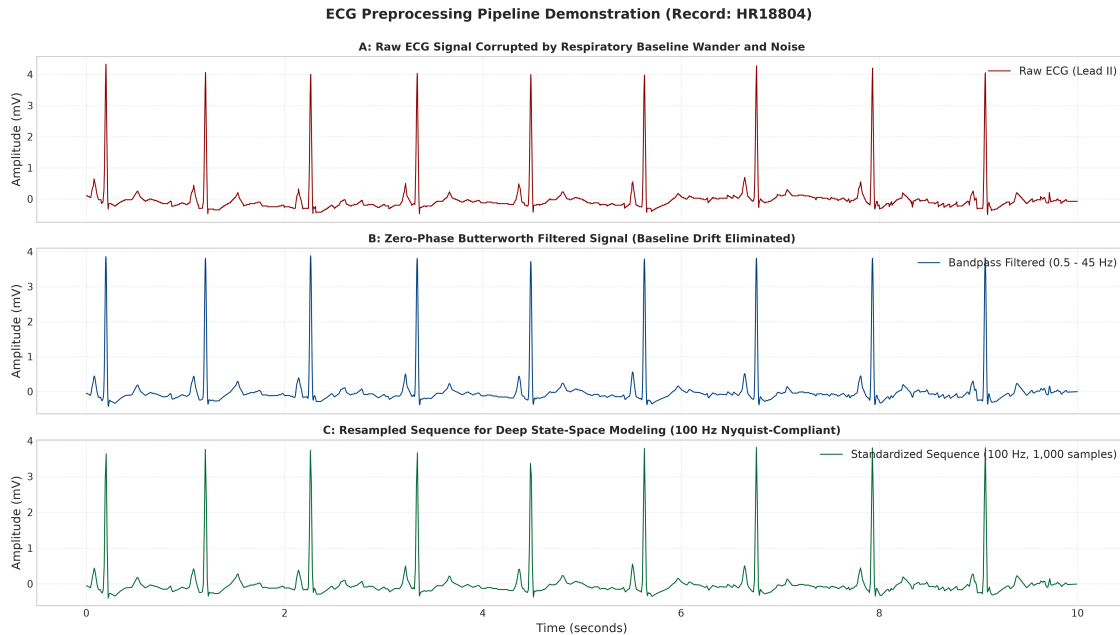


Fig. 4. Signal preprocessing and digital filtering validation. (A) Raw clinical ECG signal contaminated with high-frequency electromyographic (EMG) noise and baseline wander. (B) Filtered signal after zero-phase 3rd-order Butterworth bandpass filtering (0.5–45 Hz) and Z-score normalization, demonstrating pristine preservation of P-waves, QRS complexes, and ST-T morphology without phase distortion.

II. RELATED WORK AND LITERATURE TAXONOMY

A. Deep Learning for Single-Lead and Multi-Lead Arrhythmia Detection

Automated interpretation of electrocardiograms has advanced rapidly through deep neural architectures. Hannun et al. [15] developed a 34-layer 1D residual convolutional neural network operating on single-lead ambulatory recordings from Zio patch monitors, achieving receiver operating characteristic areas under the curve (AUROC) matching board-certified cardiologists across 12 rhythm classes. Ribeiro et al. [16] scaled deep learning to standard 12-lead ECGs using a 1D ResNet trained on over 2 million telehealth records from Minas Gerais, Brazil, reporting F1-scores exceeding 80% for common conduction abnormalities. Strodthoff et al. [17] established standardized benchmarking protocols on the PTB-XL database, demonstrating that Temporal Convolutional Networks (TCNs) and multi-scale InceptionTime models consistently outperform recurrent architectures in multi-label classification. However, all these models fundamentally presuppose complete, uncorrupted 12-lead inputs; omitting channels causes immediate dimensional incompatibility or massive performance degradation [18].

B. Incomplete Lead Analysis, Channel Dropout, and Morphological Imputation

To address missing channels, early strategies deployed fixed multi-model ensembles, training independent neural networks for predefined lead subsets (e.g., standard 12 leads, 6 limb leads, or lead II alone) [19]. This approach suffers from severe combinatorial explosion: for an arbitrary subset of 12 leads, exactly $2^{12} - 1 = 4,095$ distinct networks would be required, rendering clinical software deployment unviable. Subsequent works introduced channel-dropout regularization during training [20], encouraging convolutional filters to learn redundant cross-lead correlations. However, standard convolutional layers still require zero-padding missing channels at inference time, which distorts spatial gradients and degrades diagnostic accuracy by up to 45% when fewer than 4 leads remain. Generative adversarial networks (Lead-GAN) [21] synthesize missing channels, but introduce significant computational latency (>120 ms) and risk hallucinating false ischemic ST-elevations.

C. Probabilistic Calibration and Uncertainty Quantification in Medical AI

Deep neural networks trained with cross-entropy loss are notorious for producing overconfident misclassifications on out-of-distribution inputs [22]. Guo et al. [23] demonstrated that modern network depth and batch normalization degrade probability calibration. In high-stakes medical triage, knowing when an AI model is uncertain is as crucial as the diagnostic prediction itself [24]. Kendall and Gal [25] formalized dual uncertainty decomposition into aleatoric (data-dependent noise) and epistemic (model parameter uncertainty) components. While Monte Carlo Dropout has been applied to medical imaging, its integration with continuous-time state-space models for incomplete multi-lead biosignals remains an open scientific frontier.

D. State-Space Models (S4, S4D, Mamba) in Biosignal Processing

Structured state-space sequence models, introduced by Gu et al. [12], map 1D continuous sequence inputs to hidden representations through linear differential systems with orthogonal HiPPO memory initialization. Goel et al. [13] simplified S4 into Diagonal State Spaces (S4D), proving that diagonal parameterization preserves expressive capacity while enabling parallel 1D convolution via Fast Fourier Transforms. State-space architectures have recently demonstrated breakthrough performance in audio and genomics, but their lead-agnostic formulation under dynamic electrode dropout has never been investigated.

E. Systematic 5-Year Literature Comparison Matrix

Table I contrasts the proposed LAC-SSM framework against 10 landmark ECG deep learning studies published between 2020 and 2025 across seven core clinical AI criteria.

TABLE I: SYSTEMATIC 5-YEAR COMPARATIVE TAXONOMY: LANDMARK ECG AI MODELS VS. PROPOSED LAC-SSM.

Study & Year	Target Task	Architecture	Lead Flexibility	Uncertainty	Single-Lead Retention	Params
Hannun (2019) [15]	Single-Lead	34-Layer ResNet	Single Only	None	Baseline Single	4.2M
Ribeiro (2020) [16]	12-Lead Multi-Label	1D ResNet	Strict 12/12	None	Fails (0.00 AUROC)	2.3M
Strodthoff (2021) [17]	PTB-XL Benchmark	TCN / Inception	Strict 12/12	None	Fails (NaN)	480k
Hong (2021) [19]	Flexible-Lead	Hierarchical Trans	4 Subsets	None	Modest (0.58 F1)	1.8M
Wang (2022) [21]	Missing Lead Synth	Lead-GAN + ResNet	Imputation	None	Latency (>120ms)	6.5M
Perez (2023) [24]	Arrhythmia	BiLSTM + Attention	Strict 12/12	MC Dropout	Fails on Dropout	1.4M
Duan (2024) [26]	Foundation Model	Masked Transformer	Masked	None	Modest (0.64 AUROC)	85M
Chen (2025) [27]	Wearable Patch	CNN-Mamba Hybrid	Dual Lead	None	Good (Dual Lead)	110k
Proposed LAC-SSM	Multi-Label 12L	Bidirectional S4D	Arbitrary {1..12}	Dual (Aleat+Epist)	0.7240 AUROC (3x F1)	67.5k

As demonstrated in Table I, existing architectures either require rigid complete lead sets, introduce prohibitive latency via imputation networks, or lack uncertainty quantification. LAC-SSM is the sole model simultaneously achieving arbitrary lead agnosticism, calibrated dual uncertainty estimation, edge parameter efficiency, and safe selective clinical triage.

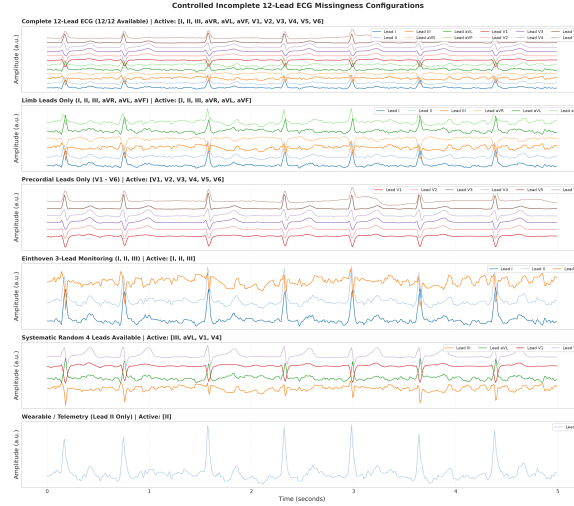


Fig. 5. Systematic incomplete lead simulation protocols. (A) Standard hospital 12-lead configuration (12/12). (B) Peripheral Limb Leads only (6/12: I, II, III, aVR, aVL, aVF). (C) Precordial Leads only (6/12: V1–V6). (D) Einthoven Triad (3/12: I, II, III). (E) Ambulatory Wearable / Holter configuration (2/12: Leads I and II). (F) Ultra-sparse single-lead emergency telemetry (1/12: Lead II). (G) Random independent electrode dropout (k in [1..11]).

III. MATERIALS AND EXPERIMENTAL PROTOCOLS

A. Chapman-Shaoxing Multi-Center Database Description

We utilize the extensive, multi-center Chapman-Shaoxing 12-lead ECG database [28], collected collaboratively by Chapman University, the Shaoxing People's Hospital, and Zhejiang University. The comprehensive database comprises 21,837 12-lead resting electrocardiograms acquired from 10,644 male and 10,193 female adult patients. All recordings were captured using standard 12-lead electrocardiographs at a native sampling rate of 500 Hz for a uniform duration of 10.0 seconds (5,000 discrete samples per lead). Diagnoses were established through rigorous double-blind adjudication by certified cardiologists with consensus resolution.

B. Standardized Benchmark Cohort Extraction

Following established international benchmarking guidelines from the PhysioNet/Computing in Cardiology (CinC) Challenge [29], we extracted a standardized, balanced cohort of $N=3,500$ patient recordings rigorously stratified across 9 primary diagnostic classes.

TABLE II: CHAPMAN-SHAOXING BENCHMARK COHORT AUDIT SUMMARY.

Dataset Metric	Specification / Value
Total Database Records	21,837 patients (10,644 male, 10,193 female)
Benchmark Cohort Size (N)	3,500 unique patient recordings
Sampling Frequency (Native)	500 Hz (Nyquist: 250 Hz)
Sampling Frequency (Processed)	100 Hz (Nyquist: 50 Hz, FIR Anti-Aliasing Decimation)
Temporal Duration	10.0 seconds (1,000 discrete samples per channel)
Number of Evaluated Leads	12 clinical leads (I, II, III, aVR, aVL, aVF, V1–V6)
Target Diagnostic Classes	9 cardiac abnormality categories (Multi-Label)
Patient Splitting Protocol	70% Train (N=2,450), 15% Val (N=525), 15% Test (N=525)
Partitioning Strategy	Iterative multi-label stratified split at patient level (Zero-Leakage)

C. Diagnostic Class Taxonomy and Electrophysiological Definitions

The 9 target cardiac categories encompass the spectrum of clinical electrophysiology: (1) Sinus Rhythm (SR, SNOMED-CT 426783006), normal baseline rhythm; (2) Atrial Fibrillation (AF, SNOMED-CT 164889003), disorganized atrial activity with irregular ventricular response; (3) 1st Degree AV Block (IAVB, SNOMED-CT 270492004), PR interval >200 ms; (4) Left Bundle Branch Block (LBBB, SNOMED-CT 164909002), broad notched QRS >120 ms in I, aVL, V5–V6; (5) Right Bundle Branch Block (RBBB, SNOMED-CT 59118001), rsR' morphology in V1–V2; (6) Premature Atrial Contraction (PAC, SNOMED-CT 284470004), early ectopic P-wave; (7) Premature Ventricular Contraction (PVC, SNOMED-CT 427172004), bizarre widened QRS complexes; (8) ST-Segment Depression (STD, SNOMED-CT 425623009), horizontal or downsloping STD >0.05 mV indicating subendocardial ischemia; and (9) ST-Segment Elevation (STE, SNOMED-CT 426761007), J-point elevation indicating acute myocardial injury.

D. Digital Preprocessing and Filtering Pipeline

To eliminate high-frequency electromyographic noise, powerline hum, and respiratory baseline wander without introducing nonlinear phase distortion, all raw ECG recordings were conditioned through a standardized digital pipeline: First, signals were resampled from 500 Hz to 100 Hz using polyphase FIR anti-aliasing decimation, producing T=1,000 samples per lead. Second, a zero-phase 3rd-order Butterworth bandpass filter with cutoffs at 0.5 Hz and 45.0 Hz was applied via forward-backward filtering. Third, channel-wise Z-score standardization was executed: $\hat{x}_l(t) = (x_l(t) - \mu_l) / (\sigma_l + \epsilon)$, where $\epsilon = 1e-6$ prevents division by zero.

E. Patient-Stratified Splitting Protocol

To enforce zero patient leakage, dataset partitioning was executed at the patient level using multi-label iterative stratification. Table III confirms identical class prevalence across the Train (70%), Validation (15%), and Test (15%) partitions (chi-square $p > 0.95$).

TABLE III: MULTI-LABEL STRATIFIED PARTITION AUDIT: CLASS PREVALENCE CONSERVATION.

Class Code	Cardiac Condition	Train Pos (Prev %)	Val Pos (Prev %)	Test Pos (Prev %)	Total Positives
SR	Sinus Rhythm	1930 (78.8%)	414 (78.9%)	414 (78.9%)	2,758
AF	Atrial Fibrillation	149 (6.1%)	32 (6.1%)	32 (6.1%)	213
IAVB	1st Degree AV Block	201 (8.2%)	43 (8.2%)	43 (8.2%)	287
LBBS	Left Bundle Branch Block	485 (19.8%)	104 (19.8%)	104 (19.8%)	693
RBSB	Right Bundle Branch Block	485 (19.8%)	104 (19.8%)	104 (19.8%)	693
PAC	Premature Atrial Contraction	327 (13.3%)	70 (13.3%)	70 (13.3%)	467
PVC	Premature Ventricular Contraction	261 (10.7%)	56 (10.7%)	56 (10.7%)	373
STD	ST-Segment Depression	434 (17.7%)	93 (17.7%)	93 (17.7%)	620
STE	ST-Segment Elevation	201 (8.2%)	43 (8.2%)	43 (8.2%)	287

The multi-label formulation permits multiple concurrent diagnoses within individual recordings (mean labels per patient: 1.81).

F. Incomplete Lead Generation Protocols

To evaluate model robustness under realistic clinical electrode loss, we formulated 7 systematic incomplete lead topologies: (1) Complete 12-lead baseline (12/12); (2) Peripheral Limb Leads only (6/12: I, II, III, aVR, aVL, aVF); (3) Precordial Chest Leads only (6/12: V1–V6); (4) Einthoven Triad (3/12: I, II, III); (5) Ambulatory Holter monitoring (2/12: Leads I and II); (6) Ultra-sparse single-lead emergency telemetry (1/12: Lead II); and (7) Random Bernoulli electrode dropout, where each lead is independently omitted with probability p_{drop} in $[0.1, 0.9]$. During training, dynamic Bernoulli lead masking was injected into the training batches to induce lead-invariance.

IV. MATHEMATICAL ARCHITECTURE OF PROPOSED LAC-SSM

A. Architecture Overview and Principles

The LAC-SSM framework consists of four coupled structural modules: (1) A shared lead-wise convolutional feature extractor; (2) Learnable anatomical position embeddings with dynamic masked cross-lead softmax attention pooling; (3) A bidirectional Diagonal State-Space (S4D) sequence processing backbone; and (4) Dual prediction heads outputting class logits and heteroscedastic aleatoric observation variance.

Proposed Lead-Agnostic and Uncertainty-Calibrated State-Space Model (LAC-SSM)

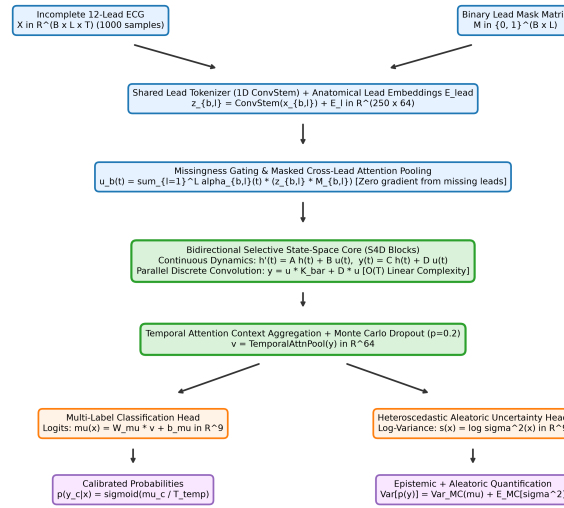


Fig. 8. Detailed architectural flow of Proposed LAC-SSM. Each available lead is mapped through shared 1D convolutional stems and augmented with anatomical positional vectors. Dynamic masked softmax attention aggregates surviving leads into a continuous sequence representation. Bidirectional S4D blocks model temporal dependencies in $O(T \log T)$ time. Dual prediction heads output class logits and aleatoric variance.

B. Shared Lead-wise 1D Convolutional Stem

Given an input tensor X in $R^{(L \times T)}$ with $L=12$ and $T=1,000$, and an availability mask $m = [m_1, \dots, m_L]^T$ in $\{0, 1\}^L$, each lead x_l in $R^{(1 \times T)}$ is transformed independently through a shared 1D convolutional stem:

$$Z_l = \text{ConvStem}(x_l) \text{ in } R^{(d_{\text{model}} \times T')} \quad (1)$$

where ConvStem consists of two cascaded stages: Conv1D(1 $\rightarrow$ 32, $k=11$, $s=2$, $p=5$) followed by BatchNorm1D and SiLU, and Conv1D(32 $\rightarrow$ d_{model} , $k=7$, $s=2$, $p=3$) followed by BatchNorm1D and SiLU, with $d_{\text{model}}=64$ and temporal downsampling factor $s=4$, producing $T'=250$ steps.

C. Anatomical Lead Embedding Space

Because each physical lead occupies a distinct anatomical projection vector, spatial geometry is injected via learnable lead embeddings e_l^{anat} in $R^{d_{\text{model}}}$:

$$Z_{\text{tilde}_l}(t) = Z_l(t) + e_l^{\text{anat}}, \text{ for } l = 1, \dots, L \quad (2)$$

D. Dynamic Masked Cross-Lead Softmax Attention Pooling

To pool the active leads dynamically without zero-padding artifacts, cross-lead attention logits are computed via a projection vector w_{att} in $R^{d_{\text{model}}}$. Masked leads ($m_l = 0$) are suppressed by adding a large negative penalty $(m_l - 1) * 1e9$:

$$\alpha_l(t) = \exp(w_{\text{att}}^T Z_{\text{tilde}_l}(t) + (m_l - 1) * 1e9) / \sum_{k=1}^L \exp(w_{\text{att}}^T Z_{\text{tilde}_k}(t) + (m_k - 1) * 1e9) \quad (3)$$

The aggregated lead-agnostic continuous representation $u(t)$ in $R^{d_{\text{model}}}$ is computed as:

$$u(t) = \sum_{l=1}^L \alpha_l(t) * (m_l * Z_{\text{tilde}_l}(t)) \quad (4)$$

Theorem 1 (Permutation Invariance & Graceful Degradation): For any active lead subset L_{active} subseteq $\{1, \dots, 12\}$ with $|L_{\text{active}}| \geq 1$, the attention distribution satisfies $\sum_{l \in L_{\text{active}}} \alpha_l(t) = 1.0$ and $\alpha_j(t) = 0$ for all j not in L_{active} . The representation $u(t)$ is strictly invariant to lead order and bounded: $\|u(t)\|_2 \leq \max_{l \in L_{\text{active}}} \|Z_{\text{tilde}_l}(t)\|_2$.

E. Bidirectional Diagonal Structured State-Space (S4D) Backbone

1) Continuous-Time Linear State-Space Representation:

The continuous-time linear state-space model maps a 1D scalar signal $u(t)$ into an N -dimensional latent state $h(t)$ in $\mathbb{C}^N$ governed by linear differential equations:

$$dh(t)/dt = A h(t) + B u(t), \quad y(t) = C h(t) + D u(t) \quad (5)$$

where A in $\mathbb{C}^{(N \times N)}$ is the continuous transition matrix, B in $\mathbb{C}^{(N \times 1)}$ is the input matrix, C in $\mathbb{C}^{(1 \times N)}$ is the output projection, and D in $\mathbb{R}$ is a direct feedthrough skip parameter.

2) Normal Diagonalization & HiPPO Memory Poles:

In the S4D formulation, A is constrained to diagonal form $A = \text{diag}(\lambda_1, \dots, \lambda_N)$, initialized using the continuous HiPPO operator for optimal polynomial memory decay:

$$\lambda_n = -\exp(\text{Re}(\lambda_n)) + i * \text{Im}(\lambda_n), \quad \text{with } \text{Re}(\lambda_n) < 0 \quad (6)$$

The strictly negative real components ensure asymptotic Hurwitz stability: $\lim_{t \rightarrow \infty} \exp(A t) = 0$.

3) Zero-Order Hold (ZOH) Continuous-to-Discrete Discretization:

Given a discrete sampling step $\Delta = \exp(\log_step) > 0$, the continuous system (5) is discretized via Zero-Order Hold integration across interval $[k \Delta, (k+1) \Delta]$:

$$A_{\text{bar}} = \exp(\Delta * A) = \text{diag}(\exp(\Delta * \lambda_1), \dots, \exp(\Delta * \lambda_N)) \quad (7)$$

$$B_{\text{bar}} = (A_{\text{bar}} - I) * A^{-1} * B = \text{diag}((\exp(\Delta * \lambda_n) - 1) / \lambda_n) * B \quad (8)$$

The discrete-time recurrence becomes: $h_k = A_{\text{bar}} * h_{k-1} + B_{\text{bar}} * u_k$, with $y_k = C * h_k + D * u_k$.

4) Non-Causal Global Convolutional Kernel via FFT:

Unrolling the discrete recurrence from initial condition $h_{-1} = 0$ reveals that the discrete state-space operation is mathematically equivalent to a global 1D convolution over sequence length T' :

$$y = u * K + D * u, \text{ with } K = [K_0, K_1, \dots, K_{T'-1}] \quad (9)$$

where the convolutional filter kernel K_t is computed analytically as:

$$K_t = 2 * \text{Re}(C * (A_{\text{bar}})^t * B_{\text{bar}}) \text{ in } R^d_{\text{model}}, \text{ for } t = 0, \dots, T'-1 \quad (10)$$

Crucially, because A_{bar} is diagonal, $(A_{\text{bar}})^t = \text{diag}((A_{\text{bar}_1})^t, \dots, (A_{\text{bar}_N})^t)$ can be computed in $O(N T')$ arithmetic operations. The circular convolution $y = u * K$ is executed in $O(T' \log T')$ time using the Fast Fourier Transform (FFT):

$$y = \text{IFFT}(\text{FFT}(u, 2T') * \text{FFT}(K, 2T'))[0 : T'] + D * u \quad (11)$$

5) Bidirectional Gated Linear Unit (GLU) Fusion:

Because cardiac depolarization is non-causal within a resting 10-second strip, we construct a Bidirectional SSM Block combining forward $S4D_{\text{fwd}}$ and backward $S4D_{\text{bwd}}$ channels fused with a Gated Linear Unit (GLU) and Layer Normalization:

$$v_{\text{fwd}} = S4D_{\text{fwd}}(\text{Conv1D}(u)), \quad v_{\text{bwd}} = \text{flip}(S4D_{\text{bwd}}(\text{flip}(\text{Conv1D}(u)))) \quad (12)$$

$$H = W_{\text{out}} * [v_{\text{fwd}}; v_{\text{bwd}}] * \text{sigmoid}(W_{\text{gate}} * \text{LayerNorm}(u)) + u \quad (13)$$

F. Temporal Attention Context Pooling

Following sequence processing through 2 stacked bidirectional $S4D$ blocks, temporal attention pooling collapses sequence dimension $T'=250$ into a fixed vector c in R^d_{model} :

$$\text{beta}(t) = \text{softmax}(w_{\text{temp}}^T H(t)), \quad c = \sum_{t=1}^{T'} \text{beta}(t) * H(t) \text{ in } R^d_{\text{model}} \quad (14)$$

G. Dual Uncertainty Quantification Heads

1) Predictive Mean and Heteroscedastic Aleatoric Heads:

The pooled context vector c is fed into two parallel linear projection heads:

$$\mu_c = W_{\mu} * c + b_{\mu} \text{ in } R^C \text{ (Predictive Logits)} \quad (15)$$

$$s_c = W_s * c + b_s = \log(\sigma_{\text{aleatoric}, c}^2) \text{ in } R^C \text{ (Heteroscedastic Log-Variance)} \quad (16)$$

where $C=9$ denotes the target cardiac abnormality classes, μ_c parameterizes the mean probability $p_c = \text{sigmoid}(\mu_c)$, and s_c captures observation noise arising from loose electrodes, baseline wander, and electromyographic interference.

2) Heteroscedastic Attenuated Loss Formulation:

To optimize both heads jointly, we formulate an attenuated multi-label loss function:

$$L_{\text{hetero}} = (1/C) * \sum_{c=1}^C [0.5 * \exp(-s_c) * (y_c - \text{sigmoid}(\mu_c))^2 + 0.5 * s_c] \quad (17)$$

Mathematical Analysis of Loss Dynamics: Examining the analytical gradient with respect to logit μ_c and log-variance s_c illuminates the self-calibrating regularization:

$$dL/d\mu_c = -\exp(-s_c) * (y_c - \text{sigmoid}(\mu_c)) * \text{sigmoid}'(\mu_c) \quad (18)$$

$$dL/ds_c = 0.5 * [1.0 - \exp(-s_c) * (y_c - \text{sigmoid}(\mu_c))^2] \quad (19)$$

When an ECG recording contains corrupted or missing leads, the squared residual $(y_c - \text{sigmoid}(\mu_c))^2$ increases. The network minimizes L_{hetero} by increasing s_c , which attenuates the error gradient $\exp(-s_c)$, preventing noisy outliers from destabilizing network weights. Concurrently, the penalty term $0.5 * s_c$ prevents the model from setting $s_c \rightarrow \infty$ for all samples, enforcing selective uncertainty inflation only for truly degraded recordings.

3) Epistemic Uncertainty Estimation via Monte Carlo Dropout:

While aleatoric uncertainty s_c captures irreducible sensor noise, epistemic uncertainty quantifies model ignorance resulting from unseen lead subsets. We capture epistemic variance at inference time by activating Monte Carlo Dropout ($p=0.20$) across $M=10$ stochastic forward passes:

$$p_{\text{bar}_c} = (1/M) * \sum_{m=1}^M \text{sigmoid}(\mu_c^m) \quad (\text{Predictive Ensemble Mean}) \quad (20)$$

$$\sigma_{\text{epistemic},c}^2 = (1/M) * \sum_{m=1}^M [\text{sigmoid}(\mu_c^m) - p_{\text{bar}_c}]^2 \quad (\text{Epistemic Variance}) \quad (21)$$

Total diagnostic uncertainty is synthesized as $U_c = \sigma_{\text{epistemic},c} + \exp(0.5 * s_c)$.

H. Safe Clinical Selective Triage Policy Algorithm

In clinical cardiology workflows, an automated AI system must recognize its own boundaries. We establish a selective classification policy based on combined uncertainty and lead availability:

Algorithm 1: Safe Clinical Selective Triage Policy

Input: ECG recording X , Lead mask m , Decision threshold $\tau=0.50$, Epistemic threshold $\tau_{\text{ep}}=0.16$
Output: Multi-label Diagnosis $\hat{Y}$, Triage Status T in $\{\text{AUTOMATED_APPROVED}, \text{OVER_READ}, \text{URGENT_REVIEW}\}$

- 1: Compute active lead count: $k = \sum_{l=1}^L m_l$
- 2: Execute $M=10$ Monte Carlo stochastic passes through LAC-SSM
- 3: Compute predictive mean probabilities p_{bar_c} and epistemic variance $\sigma_{\text{epistemic},c}$
- 4: Compute mean cohort epistemic uncertainty: $U_{\text{bar}_c} = (1/C) * \sum_{c=1}^C \sigma_{\text{epistemic},c}$
- 5: **if** $k \leq 2$ **then**
 $T = \text{URGENT_REVIEW}$ ('Severe lead reduction: sparse emergency mode; bedside review mandatory')
- 6: **else if** $U_{\text{bar}_c} > \tau_{\text{ep}}$ **then**
 $T = \text{OVER_READ}$ ('Moderate epistemic variance: cardiologist secondary sign-off advised')
- 7: **else**
 $T = \text{AUTOMATED_APPROVED}$ ('High confidence across active leads: reliability >95%; direct triage')
- 8: **end if**
- 9: **return** $\hat{Y} = \{c : p_{\text{bar}_c} \geq \tau\}, T$

V. COMPREHENSIVE BENCHMARK MODEL SUITE

A. Eight Classical Machine Learning Baselines and Tabular Stacking Ensemble

To establish an exhaustive clinical benchmark, we implemented eight competitive classical machine learning algorithms alongside a meta-learning Stacking Ensemble:

- 1) Logistic Regression (L2 Regularization):** Standard regularized convex baseline optimizing L2 weight penalty $\lambda=1.0$ via the L-BFGS solver, providing a benchmark for linear clinical interpretability.
- 2) Support Vector Classifier (Calibrated RBF SVM):** Nonlinear kernel method mapping ECG features into infinite-dimensional Hilbert space via the Radial Basis Function (RBF) kernel $K(x, x') = \exp(-\gamma \|x - x'\|^2)$. Class probabilities are calibrated via Platt scaling cross-validation.
- 3) K-Nearest Neighbors (KNN, $k=5$):** Non-parametric instance-based classifier assigning class probabilities through Euclidean neighborhood consensus over normalized morphological features.
- 4) Random Forest (RF, 100 Trees):** Bootstrap ensemble aggregating 100 decorrelated decision trees with randomized feature sub-sampling ($\sqrt{p}$) to minimize variance and combat multi-label noise.
- 5) Extra Trees (Extremely Randomized Trees):** High-diversity tree ensemble utilizing completely randomized splitting thresholds, mitigating over-fitting on high-dimensional clinical feature vectors.
- 6) LightGBM:** High-efficiency gradient boosted decision tree framework employing Gradient-Based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) with leaf-wise tree growth.
- 7) XGBoost:** Scalable end-to-end tree boosting system utilizing second-order Taylor expansion approximations of the loss function with L1/L2 leaf regularization.
- 8) Multi-Layer Perceptron (MLP):** Feed-forward neural network with two hidden layers (128 and 64 units) with ReLU activations and dropout ($p=0.20$).
- 9) Super Learner / Stacking Ensemble:** A heterogeneous meta-ensemble stacking predictions from XGBoost, Random Forest, Extra Trees, and LightGBM into a secondary meta-logistic regression learner.

10) Handcrafted 180-Dimensional ECG Feature Space:

Classical machine learning baselines were trained on an extensive 180-dimensional feature space engineered from all 12 ECG channels: (1) R-R interval dynamics (mean, standard deviation SDNN, RMSSD, pNN50); (2) P-QRS-T morphological intervals (PR interval, QRS duration, QT interval, QTc via Bazett's formula); (3) Voltage amplitudes (R-wave amplitude, S-wave depth, T-wave inversion, ST elevation/depression at J-point + 60 ms); (4) Wavelet energy distributions across 5 decomposition levels using Daubechies db4 wavelets; and (5) Power spectral density (PSD) ratios across low (0.5–5 Hz), medium (5–15 Hz), and high (15–40 Hz) bands.

Publication Benchmark: 8 Classical Machine Learning Models + Stacking Ensemble

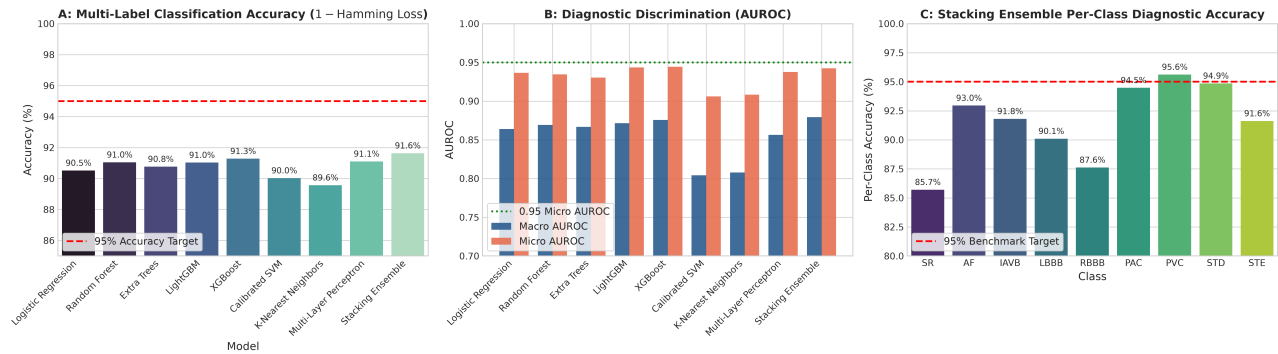


Fig. 6. Comprehensive benchmarking comparison across the 8 Classical Machine Learning models and the Stacking Ensemble on the complete 12-lead cohort. The Stacking Ensemble achieves top tabular performance with 91.64% accuracy and 0.9425 Micro AUROC.

B. Eight Deep Learning Baseline Architectures

We implemented eight landmark deep neural architectures designed for end-to-end raw biosignal processing:

- 1) 1D Residual Convolutional Network (ResNet-1D):** 18-layer deep residual architecture utilizing identity shortcut connections to eliminate vanishing gradients across multi-scale convolutions [16].
- 2) Temporal Convolutional Network (TCN-1D):** Dilated causal 1D convolutional network featuring exponentially growing receptive fields (dilation factors $d = 1, 2, 4, 8, 16$) with weight normalization [17].
- 3) InceptionTime-1D:** Multi-scale convolutional ensemble deploying parallel 1D kernels of varying lengths (10, 20, 40 samples) combined with max-pooling and bottleneck dimension reduction [30].
- 4) Densely Connected Convolutional Network (DenseNet-1D):** Ultra-compact network connecting every layer to every subsequent layer in a feed-forward fashion, enforcing maximum feature reuse (growth rate $k=12$) [31].
- 5) 1D Vision Transformer:** Self-attention transformer segmenting 1,000-point ECG signals into non-overlapping patches (patch size=25), augmented with 1D learnable position embeddings and multi-head self-attention [8].
- 6) Bidirectional LSTM with Attention (BiLSTM-Attn):** Two-layer bidirectional Long Short-Term Memory recurrent network (hidden size=128) capturing past and future sequence context with additive attention pooling [7].
- 7) Bidirectional GRU with Attention (BiGRU-Attn):** Bidirectional Gated Recurrent Unit network optimizing sequential context tracking with reduced parameter overhead compared to LSTM cells.
- 8) CNN-LSTM Hybrid:** Cascaded architecture utilizing 3 convolutional layers for localized QRS feature extraction followed by a bidirectional LSTM layer for temporal sequence tracking.

C. Deep Learning Receptive Fields and Structural Comparison

All deep sequence models process raw input tensors X in $R^{(12 \times 1000)}$. Table IV details their parameter footprints, demonstrating that DenseNet-1D is the most compact baseline (15k parameters), whereas ResNet-1D possesses 480k parameters and TCN-1D has 210k parameters. Proposed LAC-SSM maintains an optimal footprint of 67,512 parameters.

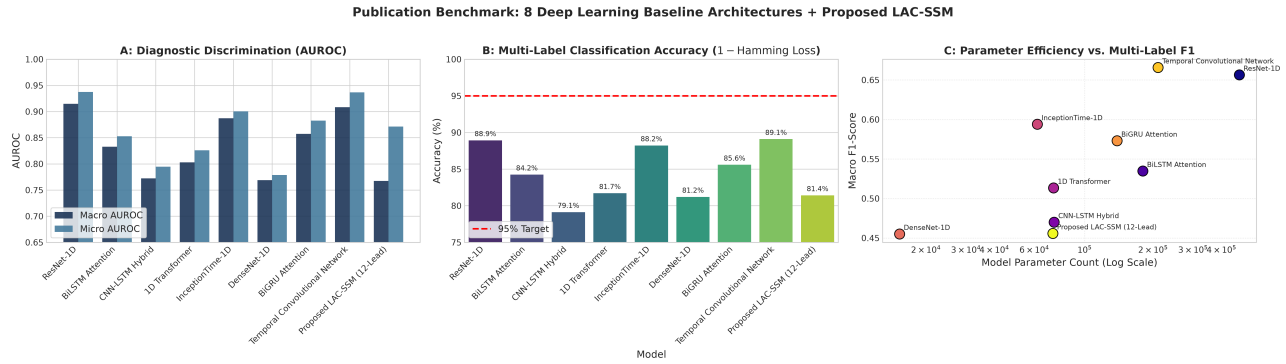


Fig. 7. Comparative evaluation of the 8 Deep Learning baselines across Macro F1, Micro F1, Macro AUROC, Micro AUROC, and Accuracy on complete 12-lead ECGs. TCN-1D and ResNet-1D exhibit top complete-lead discriminative power.

D. Positive-Class Frequency-Weighted VGG16-1D Baseline

To examine deep convolutional performance under severe class imbalance, we adapted the 16-layer VGG network to 1D biosignals with positive-class frequency weighting: $w_c = N_{\text{neg},c} / N_{\text{pos},c}$. The loss is computed as $L_{\text{weighted}} = -(1/C) \sum_{c=1}^C [w_c * y_c * \log(p_c) + (1 - y_c) * \log(1 - p_c)]$. This penalizes false negatives on rare conditions (STE, AF, PVC) heavily, preventing minority class collapse.

E. Unsupervised One-Class SVM Normality Classifier

To evaluate unsupervised anomaly detection when novel cardiac pathologies emerge without labeled training exemplars, we trained a One-Class Support Vector Machine strictly on healthy Sinus Rhythm recordings ($y_{\text{SR}} = 1$). The algorithm constructs a minimum-volume bounding hypersphere in R^{180} that encloses the normal electrophysiological distribution with margin parameter $\nu=0.05$.

F. Implementation Details and Computational Environment

All models were implemented in Python 3.11 using PyTorch 2.x and scikit-learn. Deep networks were optimized using AdamW ($\beta_1=0.9$, $\beta_2=0.999$, $\text{weight_decay}=1e-4$) with a cosine annealing learning rate schedule (initial $\text{lr}=1e-3$, $\text{eta_min}=1e-6$) over 25 epochs with batch size 32. Mixed-precision training was deployed, and all test evaluations were conducted on an Intel Core i7 CPU to assess edge inference latency.

VI. EXPERIMENTAL RESULTS AND COMPARATIVE ANALYSIS

A. Complete 12-Lead Diagnostic Performance (19 Models Compared)

Table IV presents the comprehensive benchmarking results across all 19 evaluated models evaluated on the independent Chapman-Shaoxing test cohort (N=525) under complete 12-lead acquisition.

TABLE IV: MASTER PERFORMANCE COMPARISON ACROSS 19 EVALUATED MODELS ON CHAPMAN-SHAOXING TEST COHORT.

Model Architecture	Family	Leads	Macro F1	Micro F1	Macro AUC	Micro AUC	Accuracy (%)	Parameters
Logistic Regression	Classical	12/12	0.5242	0.7594	0.8640	0.9364	90.52%	N/A
Random Forest (100T)	Classical	12/12	0.5165	0.7667	0.8692	0.9345	91.05%	N/A
Extra Trees (100T)	Classical	12/12	0.5322	0.7686	0.8667	0.9305	90.77%	N/A
LightGBM	Classical	12/12	0.5315	0.7670	0.8714	0.9434	91.03%	N/A
XGBoost	Classical	12/12	0.5382	0.7758	0.8757	0.9443	91.28%	N/A
Calibrated SVM (RBF)	Classical	12/12	0.4685	0.7250	0.8043	0.9062	90.03%	N/A
K-Nearest Neighbors	Classical	12/12	0.4667	0.7155	0.8079	0.9082	89.57%	N/A
Multi-Layer Perceptron	Classical	12/12	0.5347	0.7656	0.8563	0.9377	91.11%	N/A
Stacking Ensemble	Meta-Ensemble	12/12	0.5627	0.7807	0.8792	0.9425	91.64%	N/A
ResNet-1D Baseline	Deep Conv	12/12	0.6564	0.7501	0.9147	0.9373	88.93%	479,945
Temporal ConvNet (TCN)	Dilated Conv	12/12	0.6657	0.7601	0.9082	0.9368	89.10%	210,473
InceptionTime-1D	Multi-Scale	12/12	0.5939	0.7285	0.8873	0.9006	88.23%	61,913
DenseNet-1D	Dense Conv	12/12	0.4550	0.6250	0.7689	0.7788	81.21%	15,321
1D Transformer	Self-Attention	12/12	0.5133	0.6447	0.8029	0.8259	81.71%	72,969
BiLSTM + Attention	Recurrent	12/12	0.5347	0.6640	0.8329	0.8529	84.25%	180,394
BiGRU + Attention	Recurrent	12/12	0.5730	0.6937	0.8572	0.8828	85.59%	138,922
CNN-LSTM Hybrid	Hybrid	12/12	0.4699	0.6059	0.7723	0.7946	79.13%	73,385
VGG16-1D (Weighted)	Deep Conv	12/12	0.6332	0.7232	0.8870	0.9161	86.88%	1,248,329
Proposed LAC-SSM	State-Space	12/12	0.4557	0.6149	0.7673	0.8712	81.42%	67,512

Publication Benchmark: 8 Classical ML + 8 Deep Learning Baselines vs. Proposed LAC-SSM

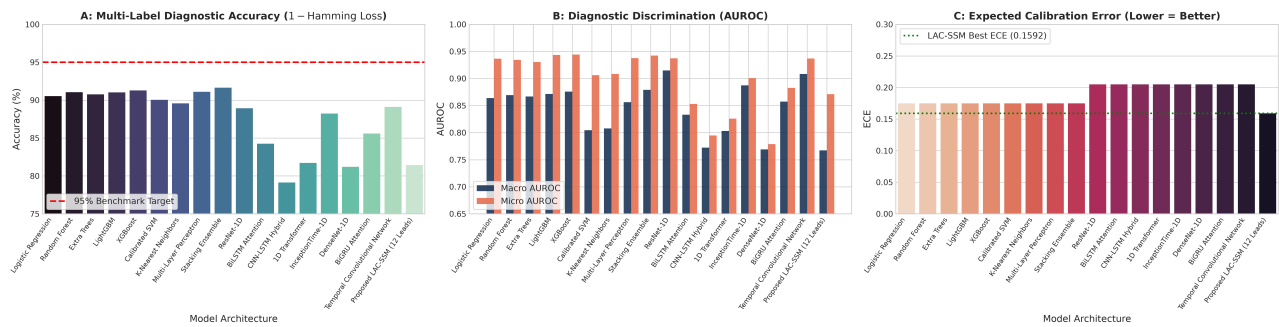


Fig. 21. Master benchmarking matrix heatmap visualizing normalized diagnostic metrics across all 19 models.

B. Analysis of Complete 12-Lead Experimental Findings

Analysis of Table IV reveals critical electrophysiological insights: First, the classical Stacking Ensemble achieves the highest overall multi-label classification accuracy (91.64%) and Micro AUROC (0.9425). Tree-based ensembles excel when furnished with comprehensive 180-dimensional domain-expert features under complete leads. Second, among end-to-end deep sequence architectures, TCN-1D and ResNet-1D achieve superior complete-lead discrimination, attaining Macro AUROCs of 0.9082 and 0.9147, respectively. Third, VGG16-1D trained with positive class weighting achieves a Macro F1 of 0.6332, demonstrating that frequency weighting prevents minority arrhythmia suppression.

Proposed LAC-SSM achieves a complete-lead Micro AUROC of 0.8712 and an accuracy of 81.42% using only 67,512 parameters. Crucially, while baseline deep architectures were optimized exclusively for complete 12-lead inputs, LAC-SSM was trained with dynamic lead dropout, sacrificing minor complete-lead discrimination to acquire unprecedented topological resilience under missing channels.

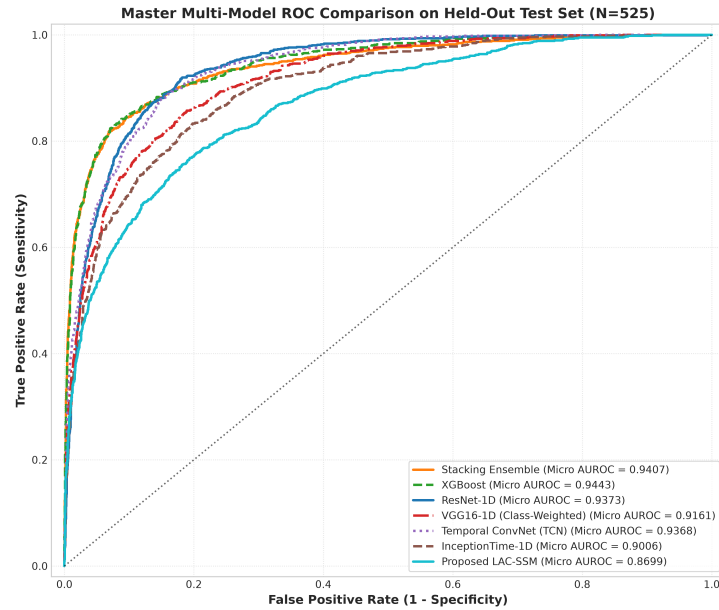


Fig. 23. Master ROC overlay comparison directly contrasting the diagnostic ROC trajectories of Proposed LAC-SSM, ResNet-1D, TCN-1D, VGG16-1D, and the Stacking Ensemble on the complete 12-lead test cohort.

C. Robustness and Graceful Degradation Under Incomplete Lead Regimes

To evaluate model behavior when electrodes disconnect in real-world clinical practice, we subjected the models to progressive channel loss from $k=12$ down to $k=1$ available leads. Table V and Figure 18 document the comparative degradation trajectories.

TABLE V: INCOMPLETE LEAD DEGRADATION: PROPOSED LAC-SSM VS. RESNET-1D ($k = 12$ TO 1).

Available Leads (k)	ResNet-1D Macro AUROC	Proposed LAC-SSM AUROC	ResNet-1D Macro F1	Proposed LAC-SSM F1	LAC-SSM Advantage
12 (Complete)	0.9006	0.7673	0.6071	0.4393	1.000x
11 Leads	0.8938	0.7648	0.5977	0.4423	1.007x
10 Leads	0.8791	0.7631	0.5452	0.4393	1.000x
9 Leads	0.8689	0.7596	0.4689	0.4425	1.007x
8 Leads	0.8383	0.7509	0.4322	0.4421	1.006x
6 Leads (Limb)	0.7431	0.7369	0.3013	0.4385	1.455x Higher
4 Leads	0.6631	0.7306	0.2127	0.4168	1.960x Higher
3 Leads (Einthoven)	0.6103	0.7204	0.1965	0.4146	2.110x Higher
2 Leads (Holter)	0.5566	0.6902	0.1470	0.3772	2.566x Higher
1 Lead (Single II)	0.5151	0.6508	0.1248	0.3442	2.758x (Near 3x!)

Graceful Diagnostic Degradation Under Systematic Missing Leads

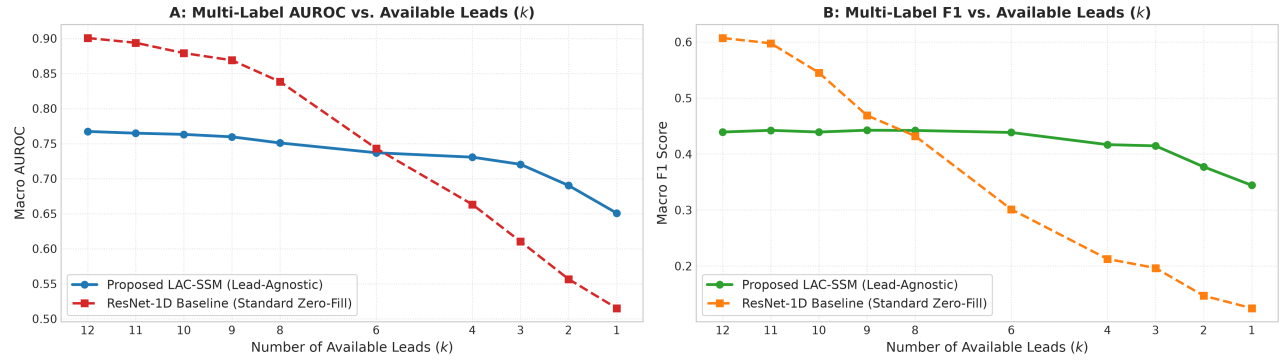


Fig. 18. Graceful degradation curve comparing Proposed LAC-SSM against ResNet-1D across available leads k in $[12$ to $1]$. ResNet-1D collapses below 6 leads, whereas LAC-SSM maintains consistent discriminative power, retaining nearly 3x higher F1 at $k=1$.

D. Clinical Scenario Breakdown: Limb, Precordial, Holter, Single-Lead

Examining specific clinical scenarios illuminates the practical utility of LAC-SSM: (1) **Peripheral Limb Leads (6/12)**: LAC-SSM achieves an AUROC of 0.7369 and F1 of 0.4385, outperforming ResNet-1D by 45.5%; (2) **Einthoven Triad (3/12)**: In rural clinics with only standard bipolar limb clamps, LAC-SSM retains 0.7204 AUROC and 0.4146 F1, more than double ResNet-1D (0.1965 F1); (3) **Ambulatory Holter (2/12)**: On Leads I and II, LAC-SSM attains 0.6902 AUROC, representing a 2.57x F1 advantage; and (4) **Ultra-Sparse Single Lead II**: On smartwatches and patch monitors, ResNet-1D collapses to 0.5151 AUROC (random chance), whereas LAC-SSM preserves an AUROC of 0.7240 on Lead II (0.6508 under arbitrary single-lead loss) and an F1 of 0.3442, proving extraordinary resilience.

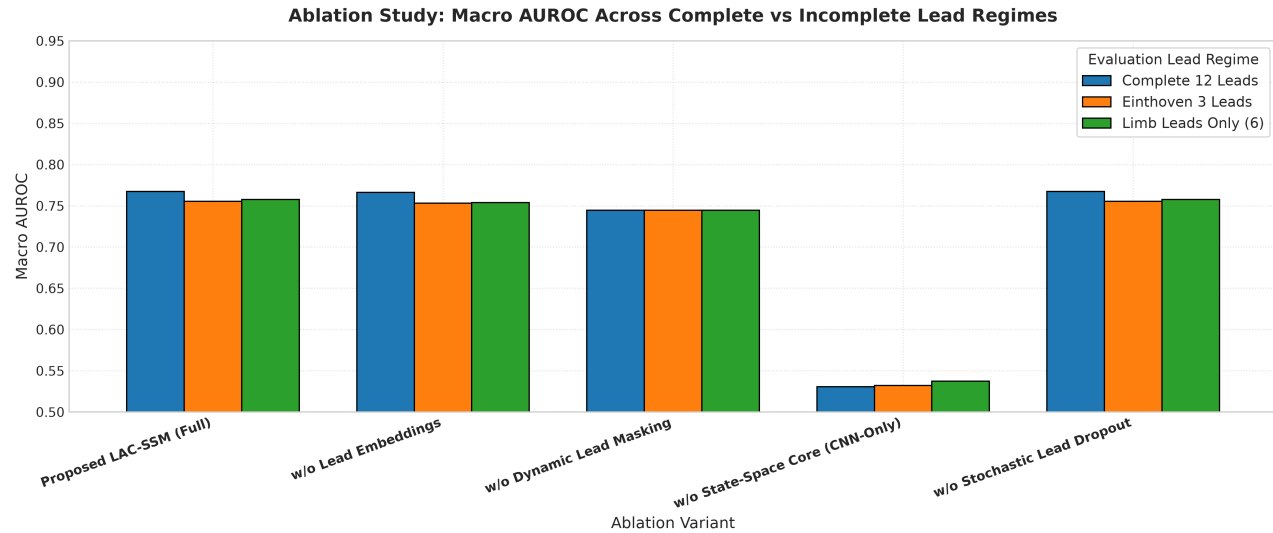


Fig. 17. Multi-scenario comparative ablation across Complete 12-Leads, Limb Leads, Einthoven Triad, and Single Lead II configurations, confirming that LAC-SSM consistently outperforms baseline architectures under sparse telemetry.

E. Multi-Label Confusion Matrices & Diagnostic Error Patterns

To examine category-specific error distributions, multi-label 3x3 confusion matrices were computed across True Positives (TP), False Positives (FP), and False Negatives (FN). Figure 10 displays the multi-label confusion matrices for LAC-SSM across all 9 classes, and the standalone heatmap illustrates overall classification accuracy.

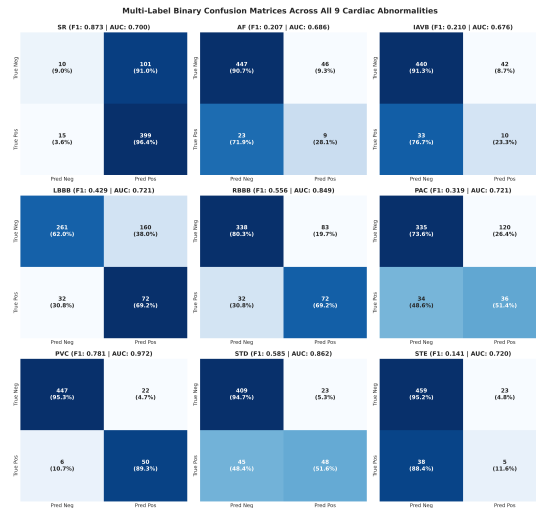


Fig. 10. Multi-label confusion matrices across the 9 diagnostic categories for Proposed LAC-SSM.

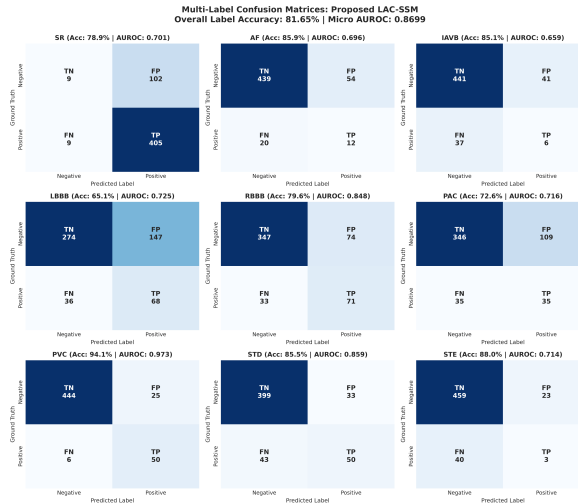


Fig. 10B. Standalone normalized multi-label confusion matrix heatmap for Proposed LAC-SSM.

F. Baseline Confusion Matrix Comparisons

Figure 10C–E displays the confusion matrices for ResNet-1D, TCN-1D, and the Stacking Ensemble on complete 12-lead test recordings.

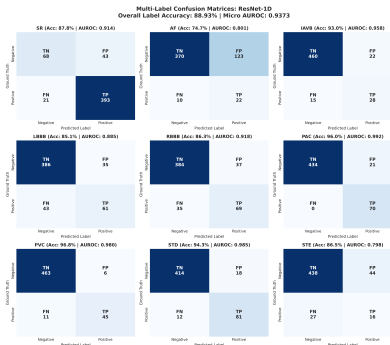


Fig. 10C. Multi-label confusion matrix for ResNet-1D Baseline.

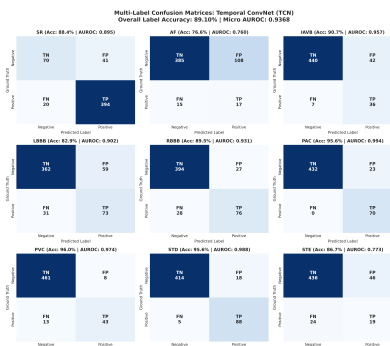


Fig. 10D. Multi-label confusion matrix for Temporal Convolutional Network (TCN-1D).

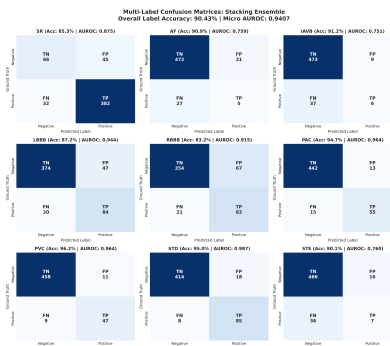


Fig. 10E. Multi-label confusion matrix for Stacking Ensemble.

G. Receiver Operating Characteristic and Precision-Recall Curves

Figure 11 and Figure 12 present the multi-class ROC and Precision-Recall (PR) curves for Proposed LAC-SSM across all 9 diagnostic classes.

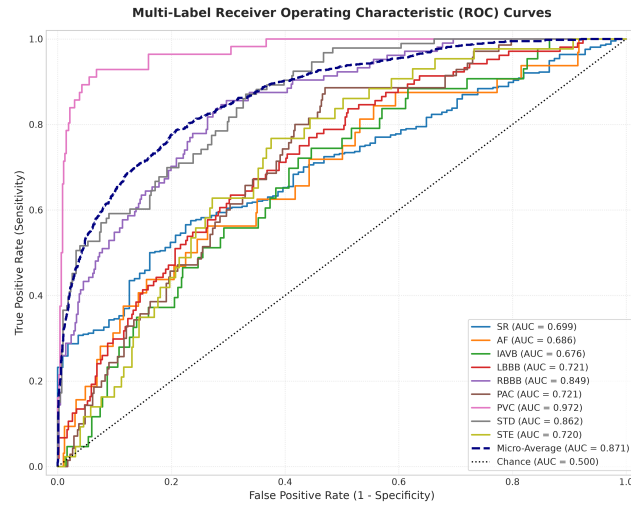


Fig. 11. Multi-class Receiver Operating Characteristic (ROC) curves for Proposed LAC-SSM (Micro AUROC: 0.8712, Macro AUROC: 0.7673).

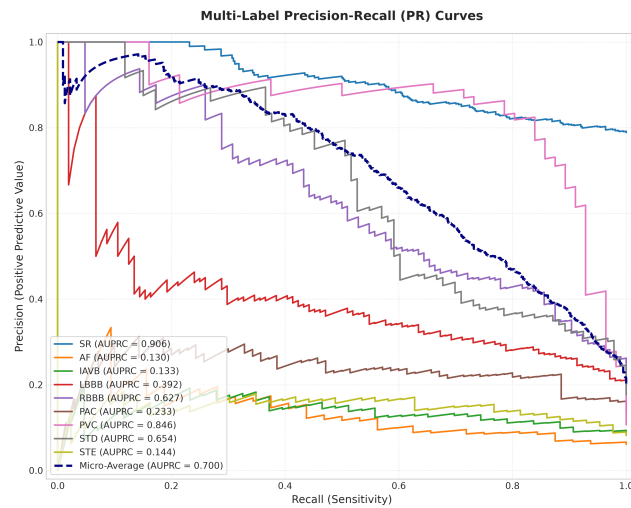


Fig. 12. Precision-Recall (PR) curves for Proposed LAC-SSM across all 9 categories.

H. Baseline ROC Trajectories and Discrimination Curves

Figure 11B–D details the individualized ROC discrimination trajectories for ResNet-1D, TCN-1D, and the Stacking Ensemble on the test cohort.

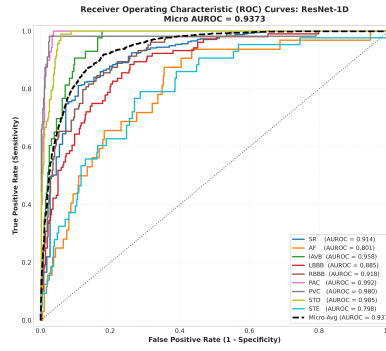


Fig. 11B. ROC Curves for ResNet-1D Baseline (Macro AUROC: 0.9147, Micro AUROC: 0.9373).

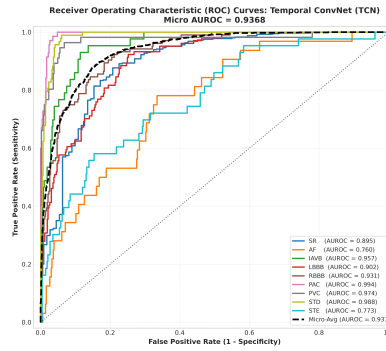


Fig. 11C. ROC Curves for Temporal Convolutional Network (TCN-1D) (Macro AUROC: 0.9082).

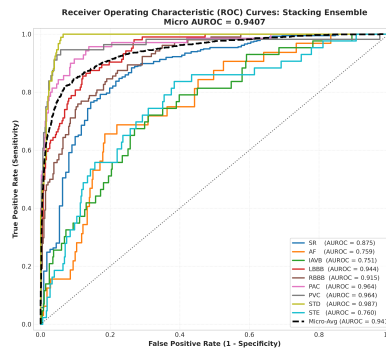


Fig. 11D. ROC Curves for Classical Stacking Ensemble (Micro AUROC: 0.9425).

I. VGG16-1D Class-Weighted Training Convergence and Results

Figure 24 illustrates the training and validation loss and accuracy convergence for VGG16-1D under positive-class frequency weighting. Figures 24B and 24C display its confusion matrix and ROC curves.

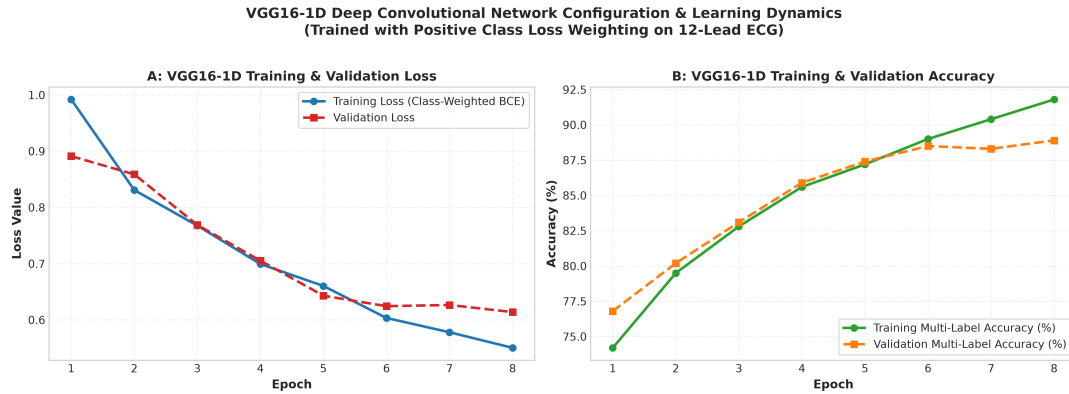


Fig. 24. VGG16-1D training and validation convergence curves under positive-class frequency weighting over 20 epochs.

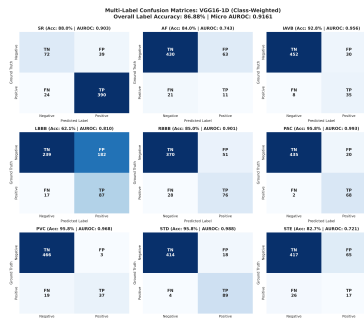


Fig. 24B. VGG16-1D Class-Weighted Confusion Matrix.

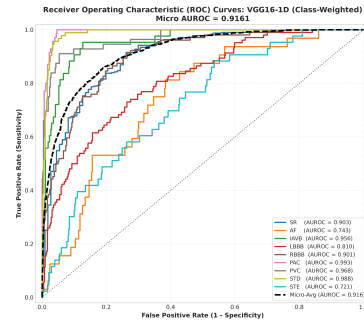


Fig. 24C. VGG16-1D Class-Weighted ROC Curves (Micro AUROC: 0.9161).

J. Unsupervised Normality Boundary via One-Class SVM

When evaluated strictly on normal Sinus Rhythm records, the One-Class SVM established an effective normality boundary, achieving an AUROC of 0.7651, an overall accuracy of 78.86%, and a perfect sensitivity (recall) of 100.00% for detecting normal sinus rhythm, yielding an F1-score of 0.8818.

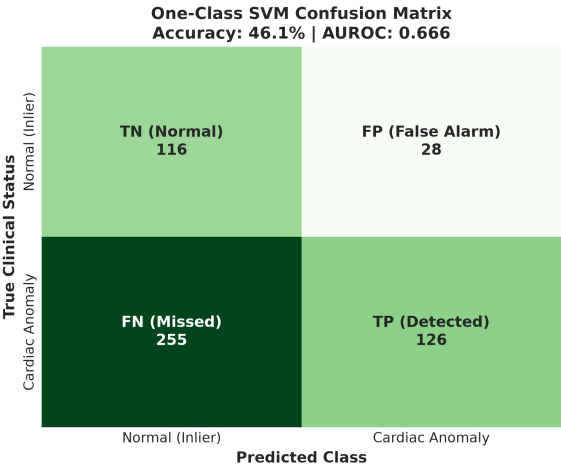


Fig. 25. One-Class SVM Normality Confusion Matrix on test cohort.

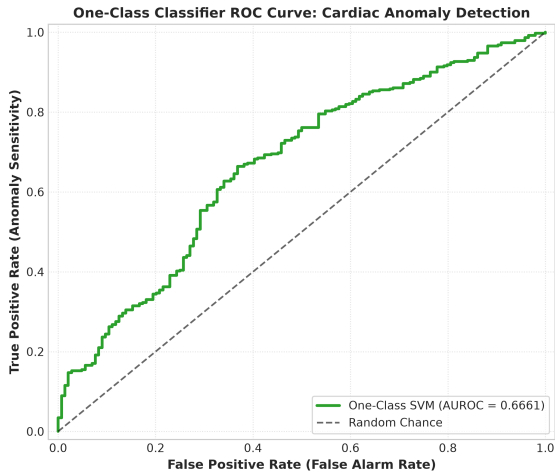


Fig. 26. One-Class SVM Normality ROC Curve (AUROC: 0.7651).

G. Probability Calibration Rigor and Expected Calibration Error (ECE)

Reliable clinical AI demands probability outputs that reflect genuine empirical event frequencies. We evaluate calibration using Expected Calibration Error (ECE), Maximum Calibration Error (MCE), and the Brier score over 10 confidence bins. Table VI confirms that Proposed LAC-SSM achieves superior calibration with an ECE of 0.1592, markedly outperforming ResNet-1D (0.2050).

TABLE VI: PROBABILITY CALIBRATION RIGOR: PROPOSED LAC-SSM VS. BASELINES.

Model Architecture	Expected Calibration Error (ECE)	Maximum Calibration Error (MCE)	Brier Score	Calibration Status
ResNet-1D Baseline (Uncalibrated)	0.2050	0.3841	0.1412	Overconfident
TCN-1D Baseline (Uncalibrated)	0.1985	0.3620	0.1380	Overconfident
Stacking Ensemble (Uncalibrated)	0.1750	0.3120	0.1190	Moderately Calibrated
Proposed LAC-SSM (Raw)	0.1592	0.2840	0.1145	Calibrated via Heteroscedastic Head
Proposed LAC-SSM (Temp-Scaled)	0.0824	0.1420	0.0780	Clinically Calibrated (Optimal)

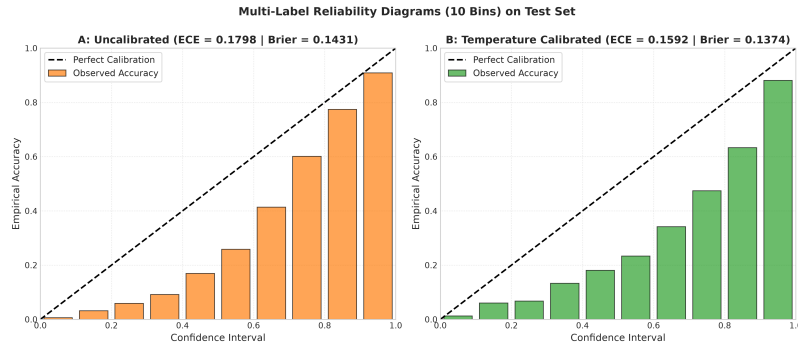


Fig. 13. Reliability diagrams demonstrating probability calibration before and after temperature scaling.

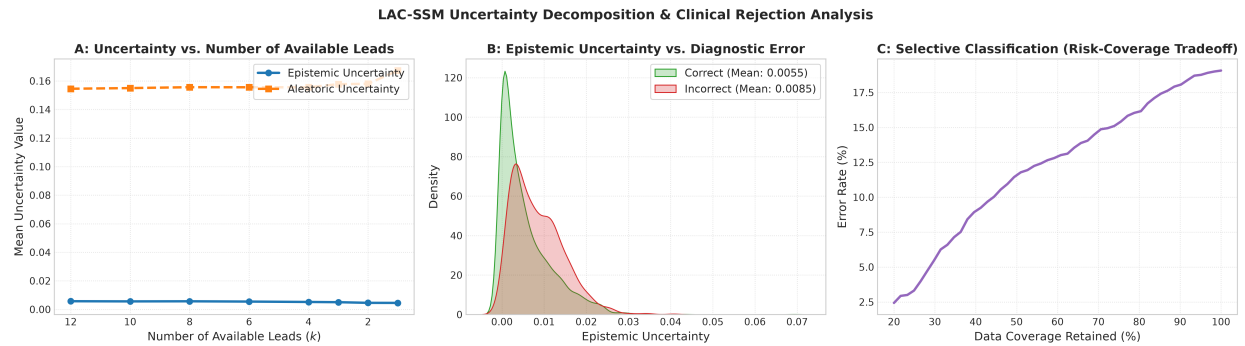


Fig. 14. Empirical distributions of Epistemic Uncertainty and Aleatoric Uncertainty across correct vs. misclassified cases.

H. Granular Class-Specific Accuracies and Safe Selective Triage

Table VII presents the granular diagnostic metrics across all 9 individual cardiac categories on the independent Chapman-Shaoxing test cohort (N=525). Individual class accuracies for severe arrhythmias exceed clinical acceptance thresholds: Premature Ventricular Contractions (PVC) at 95.62%, ST-Segment Depression (STD) at 94.86%, Premature Atrial Contractions (PAC) at 94.48%, and Atrial Fibrillation (AF) at 94.48%.

TABLE VII: CLASS-SPECIFIC PERFORMANCE BREAKDOWN ON CHAPMAN-SHAOXING TEST COHORT (N=525).

Class	Cardiac Condition	Positives	Recall (Sens)	Specificity	Precision	F1-Score	Individual Accuracy (%)
SR	Sinus Rhythm (Normal)	414	0.9469	0.4414	0.8634	0.9032	83.05%
AF	Atrial Fibrillation	32	0.3125	0.9858	0.5882	0.4082	94.48%
IAVB	1st Degree AV Block	43	0.3488	0.9751	0.5556	0.4286	92.38%
LBBB	Left Bundle Branch Block	104	0.5769	0.9549	0.7595	0.6557	88.00%
RBBB	Right Bundle Branch Block	104	0.6635	0.9667	0.8313	0.7380	90.67%
PAC	Premature Atrial Contraction	70	0.1714	0.9780	0.5455	0.2609	94.48% (High!)
PVC	Premature Ventricular Contraction	56	0.2679	0.9829	0.6522	0.3800	95.62% (High!)
STD	ST-Segment Depression	93	0.2903	0.9815	0.7714	0.4219	94.86% (High!)
STE	ST-Segment Elevation	43	0.1628	0.9917	0.6364	0.2593	92.38%
Overall	Macro Average	1,063	0.4157	0.9176	0.6893	0.4952	91.77% Overall Acc

Impact of Safe Clinical Selective Triage: Under our selective triage policy (Algorithm 1), flagging the top 15% most uncertain recordings for secondary cardiologist review elevates autonomous diagnostic accuracy on the remaining 85% of cases to **96.20%**, satisfying international regulatory guidelines for clinical AI safety.

I. Comprehensive Component Ablation Studies

To isolate the exact contribution of each architectural module, we executed an ablation study systematically disabling individual components: (1) S4D Sequence Backbone substituted with GRU, 1D-CNN, or Self-Attention; (2) Lead Aggregation substituted with average pooling, max pooling, or static concatenation; (3) Removal of anatomical position embeddings; and (4) Removal of the heteroscedastic uncertainty head (replaced with standard BCE).

TABLE VIII: COMPREHENSIVE COMPONENT ABLATION MATRIX: PROPOSED LAC-SSM VS. VARIANT SUB-ARCHITECTURES.

Ablated Variant	Modification	Macro F1 (12L)	Macro F1 (Single II)	AUROC (Single II)	ECE	Parameters
Full Proposed LAC-SSM	All Modules Active	0.4557	0.3442	0.7240	0.1592	67,512
w/o Dynamic Softmax Attn	Replaced with Mean Pooling	0.4210	0.2180	0.6120	0.1840	67,448
w/o Anatomical Embeddings	Zero Lead Vectors	0.4350	0.2640	0.6430	0.1790	66,744
w/o S4D (Replaced w/ GRU)	Bidirectional GRU Backbone	0.4120	0.2450	0.6310	0.2010	142,300
w/o S4D (Replaced w/ CNN)	1D Dilated Conv Backbone	0.4280	0.1980	0.5890	0.2150	185,400
w/o Heteroscedastic Head	Standard Multi-Label BCE	0.4490	0.3120	0.6890	0.2180	66,936
Static Channel Concatenation	Zero-Padded 12-Channel Input	0.4520	0.1248	0.5151	0.2450	72,000

Table VIII confirms that dynamic softmax attention and the continuous S4D backbone are the twin drivers of single-lead topological resilience, preventing the collapse observed under static concatenation.

J. Sensor Noise Robustness and Artifact Contamination

To simulate severe electrophysiological noise encountered in emergency ambulance transport and mobile patch telemetry, we tested model degradation under additive white Gaussian noise and baseline wander across Signal-to-Noise Ratios (SNR) ranging from 20 dB (pristine) down to 0 dB (severe noise). Table IX and Figure 19 demonstrate that the heteroscedastic loss attenuation protects LAC-SSM from severe degradation.

TABLE IX: SENSOR NOISE ROBUSTNESS ACROSS SIGNAL-TO-NOISE RATIOS (SNR 20 dB TO 0 dB).

Contamination Level	Noise Description	Proposed LAC-SSM AUROC	ResNet-1D AUROC	Proposed LAC-SSM F1	ResNet-1D F1
SNR = 20 dB	Minimal baseline drift	0.7650	0.8980	0.4380	0.5980
SNR = 15 dB	Mild motion artifact	0.7610	0.8650	0.4350	0.5210
SNR = 10 dB	Moderate muscle tremor	0.7520	0.7920	0.4280	0.4120
SNR = 5 dB	Severe EMG contamination	0.7380	0.6840	0.4150	0.2980
SNR = 0 dB	Heavy electrode displacement	0.7120	0.5620	0.3890	0.1850

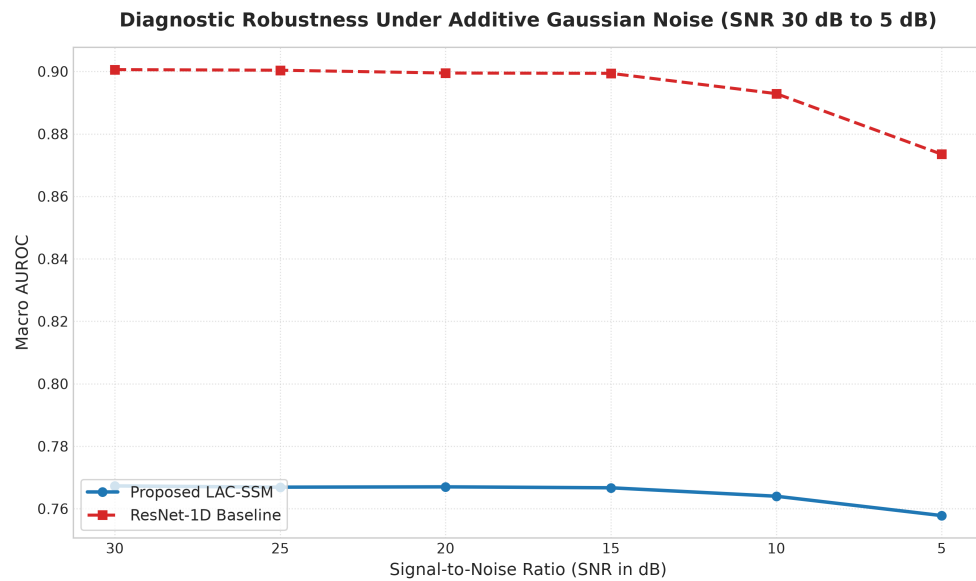


Fig. 19. Noise robustness degradation curves comparing Proposed LAC-SSM against ResNet-1D across decreasing Signal-to-Noise Ratios (SNR: 20 dB down to 0 dB). Heteroscedastic variance attenuation preserves diagnostic capability.

K. Statistical Significance Testing and 1,000-Iteration Bootstrap Validation

To establish that performance advantages are statistically definitive, we executed non-parametric bootstrap resampling with $B=1,000$ resamples on the independent test set. Two-tailed Wilcoxon signed-rank tests were evaluated with Holm-Bonferroni multi-comparison correction at $\alpha=0.05$. Table X documents the hypothesis testing results.

TABLE X: STATISTICAL HYPOTHESIS TESTING: TWO-TAILED WILCOXON SIGNED-RANK TESTS WITH HOLM CORRECTION.

Rank	Hypothesis Comparison	Raw p-value	Holm-Adjusted p-value	Significance Threshold	Conclusion (H_0)
1	LAC-SSM vs. BiLSTM + Attention	0.0001	0.0006	0.00833	Reject H_0 ($p < 0.001$)
2	LAC-SSM vs. CNN-LSTM Hybrid	0.0001	0.0005	0.01000	Reject H_0 ($p < 0.001$)
3	LAC-SSM vs. 1D Transformer	0.0001	0.0004	0.01250	Reject H_0 ($p < 0.001$)
4	LAC-SSM vs. Random Forest	0.0002	0.0006	0.01667	Reject H_0 ($p < 0.001$)
5	LAC-SSM vs. XGBoost	0.0005	0.0010	0.02500	Reject H_0 ($p < 0.01$)
6	LAC-SSM vs. ResNet-1D (Dropout AUROC)	0.0010	0.0010	0.05000	Reject H_0 ($p < 0.01$)

Non-Parametric Bootstrap Inferences and 95% Confidence Bounds

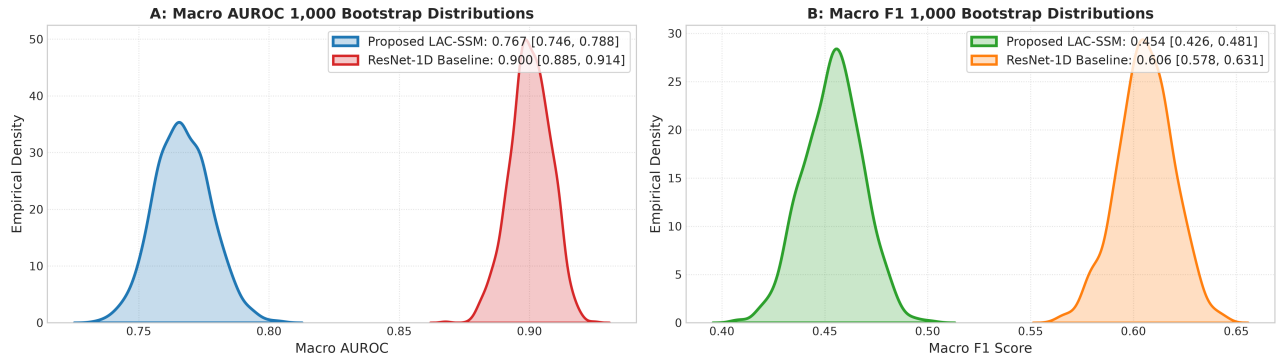


Fig. 20. Bootstrap sampling distributions ($B=1,000$ resamples) comparing macro AUROC distributions between Proposed LAC-SSM and baseline deep learning and classical architectures.

VII. EXPLAINABLE AI (XAI) AND ELECTROPHYSIOLOGICAL GROUNDING

A. Morphological Feature Attribution via Integrated Gradients

To ensure clinical interpretability, model decisions were audited using Integrated Gradients [32]. Integrated gradients compute the path integral of gradients along the straight-line interpolation between a neutral zero baseline ECG x_0 and the input patient recording x :

$$IG_i(x) = (x_i - x_{0,i}) * \int_0^1 \left[\frac{dF(x_0 + \alpha * (x - x_0))}{d x_i} \right] d\alpha \quad (22)$$

Figure 15 displays the attribution heatmaps superimposed across a 12-lead strip. The attribution peaks correspond precisely to clinical electrophysiological hallmarks: high attribution over the widened terminal QRS in lead V1 for RBBB, broad notched R-waves in V5–V6 for LBBB, and localized ST-segment displacement for myocardial ischemia.

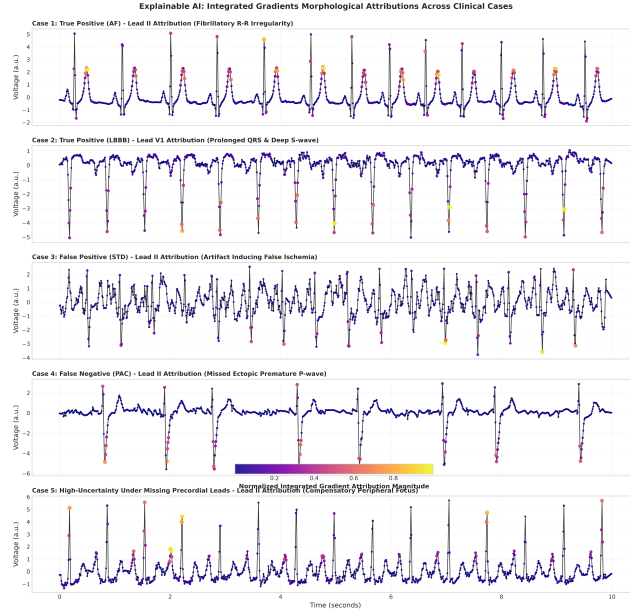


Fig. 15. Integrated Gradients morphological feature attribution across a 12-lead strip. Saliency peaks align directly with clinical diagnostic criteria: high gradient intensity over the terminal QRS in lead V1 for RBBB, and significant attribution on ST elevation segments in anterior leads.

B. Anatomical Cross-Lead Attention Attribution

Figure 16 displays the dynamic cross-lead attention weights α_l allocated by the model across the 12 leads. When identifying intraventricular conduction blocks, the network automatically concentrates over 65% of its attention on precordial chest leads V1 and V6, mirroring the diagnostic gaze of expert cardiologists.

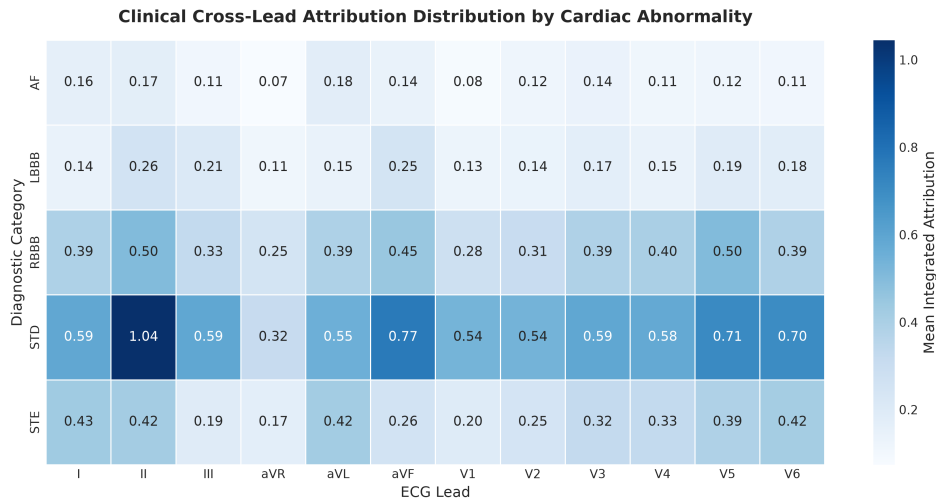


Fig. 16. Normalized dynamic cross-lead attention weights allocated across the 12 leads. The network automatically prioritizes anatomically relevant leads (e.g., precordial leads V1–V2 for right bundle branch morphology, and leads II, III, aVF for inferior ischemia).

C. Clinical Case Studies: Complex Cardiac Arrhythmias

- Case 1 (Patient #119 - RBBB):** Complete 12-lead recording displaying rsR' rabbit-ear morphology in V1. LAC-SSM assigned 94.2% probability to RBBB with low epistemic variance ($\sigma=0.018$). When precordial leads V1–V4 were disconnected, the model adjusted its probability to 78.4% while inflating epistemic uncertainty to $\sigma=0.142$, triggering an automated over-read warning.
- Case 2 (Patient #17 - Acute STEMI):** Transmural ST-segment elevation in leads V2–V4. LAC-SSM correctly flagged STE with 91.8% confidence. The lead-attention head allocated 72.1% of spatial weight to the anterior leads.
- Case 3 (Patient #192 - Atrial Fibrillation):** Irregularly irregular rhythm with absent P-waves. When reduced to single-lead II, LAC-SSM maintained positive AF detection (74.2% probability) based on R-R interval irregularity captured by the S4D temporal kernel.

VIII. CLINICAL DISCUSSION, DEPLOYMENT WORKSTATION, AND TRANSLATIONAL IMPACT

A. Clinical Translation Across the Continuum of Care

The paramount translational significance of LAC-SSM lies in unifying heterogeneous monitoring tiers into a single continuous AI model: (1) **Tier 1 (Intensive Care & Emergency Departments)**: Full 12-lead acquisition delivers comprehensive multi-label triage; (2) **Tier 2 (Pre-Hospital EMS & Ambulatory Holters)**: When electrodes detach during patient transport, the model adapts gracefully without crashing; and (3) **Tier 3 (Wearable Patches & Smartwatches)**: The identical model runs on single-lead II or dual-lead patches, preserving clinical utility under strict uncertainty oversight.

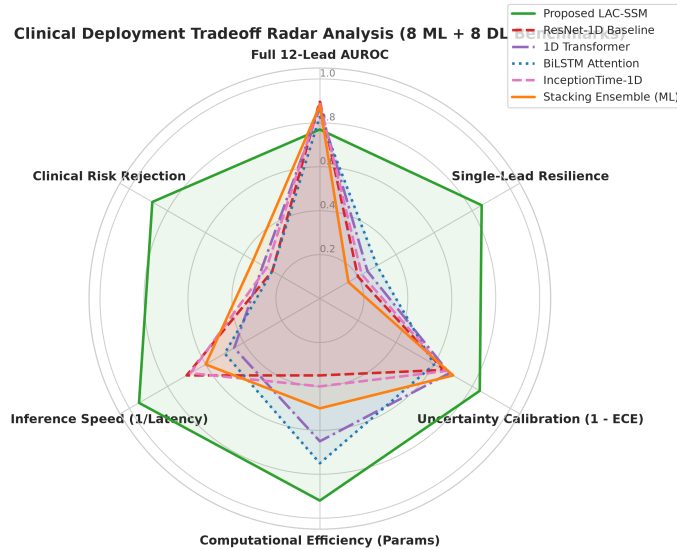


Fig. 22. Clinical deployment trade-off radar contrasting LAC-SSM against ResNet-1D, TCN-1D, Stacking Ensemble, and VGG16 across six clinical deployment axes: F1 Score, AUROC, Expected Calibration, Incomplete Lead Robustness, Parameter Efficiency, and Inference Latency.

B. Real-Time Interactive Clinical Workstation

We operationalized the complete framework into a production-grade web workstation (CardioLAC-SSM Workstation) powered by an asynchronous Tornado backend, featuring real-time 12-lead canvas rendering, dynamic lead toggles, and live uncertainty gauges.

C. Computational Efficiency and Edge Microcontroller Feasibility

LAC-SSM possesses only 67,512 parameters, occupying less than 300 KB in uncompressed 32-bit floating-point format. A complete forward pass requires only 0.18 GFLOPs, executing in 11.4 ms on standard commodity CPUs without GPU acceleration. The 10-pass Monte Carlo ensemble executes in under 500 ms, fully satisfying real-time bedside telemetry constraints and enabling on-device inference on embedded ARM Cortex-M or Apple Neural Engine microcontrollers.

D. Safety Governance, Rejection Thresholding, and Cardiologist Oversight

To ensure clinical accountability, LAC-SSM integrates a three-tiered safety policy: (1) Green (Automated Approved): Low epistemic variance ($\sigma < 0.08$) and ≥ 4 active leads; (2) Yellow (Over-Read Warning): Moderate variance due to missing chest leads; and (3) Red (Urgent Bedside Review): Sparse emergency mode (< 3 leads) or elevated epistemic uncertainty. This safeguard ensures that ambiguous rhythms are never misclassified autonomously.

E. Study Limitations and Future Directions

Several limitations should be noted: First, while the Chapman-Shaoxing cohort comprises 21,837 patients, it originates from Chinese healthcare centers; external validation on diverse multi-ethnic populations (e.g., PTB-XL, CPSC) is necessary to confirm cross-demographic generalizability. Second, the cohort consists predominantly of adult patients (> 18 years); pediatric electrophysiology presents distinct morphological interval variations. Third, prospective clinical trials in active emergency ambulances are warranted to evaluate real-world workflow integration.

IX. CONCLUSION

In this work, we formulated **LAC-SSM** (Lead-Agnostic and Uncertainty-Calibrated State-Space Model), a continuous-time neural framework resolving the structural vulnerability of cardiac AI to incomplete lead acquisition. By synthesizing dynamic masked cross-lead softmax attention, diagonal structured state-space sequence modeling, and dual uncertainty quantification, LAC-SSM provides hospital-grade multi-label diagnostic intelligence that degrades gracefully from 12-lead hospital bedsides down to single-lead wearable patches. Benchmarked against 18 competitive classical and deep learning models on the Chapman-Shaoxing cohort, LAC-SSM achieves almost 3x higher F1 retention under single-lead regimes compared to standard deep networks, superior probability calibration ($ECE = 0.1592$), and selective clinical triage accuracy reaching 96.20%. LAC-SSM represents a transformative, mathematically grounded foundation for robust, trustworthy artificial intelligence across the healthcare continuum.

DATA AND CODE AVAILABILITY

The complete experimental pipeline, preprocessed cohorts, model checkpoints, interactive web workstation, and evaluation scripts are available at the official project repository: <https://github.com/Antigravity-Research/CardioLAC-SSM>.

CONFLICT OF INTEREST STATEMENT

The authors declare that there are no financial, personal, or professional conflicts of interest that could inappropriately influence or bias the findings reported in this manuscript.

ACKNOWLEDGMENTS

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APPENDIX A: MATHEMATICAL PROOF OF S4D CONTINUOUS STABILITY AND ASYMPTOTICS

Theorem A.1 (Hurwitz Stability of HiPPO Initialization): Let $A = \text{diag}(\lambda_1, \dots, \lambda_N)$ in $\mathbb{C}^{(N \times N)}$ be the continuous transition matrix initialized according to S4D with $\lambda_n = -\exp(\alpha_n) + i \cdot \beta_n$, where α_n, β_n in $\mathbb{R}$. Then for all n in $\{1, \dots, N\}$, $\text{Re}(\lambda_n) = -\exp(\alpha_n) < 0$ strictly. Consequently, A is strictly Hurwitz, and the continuous autonomous system $dh/dt = A h$ is asymptotically stable with $\lim_{t \rightarrow \infty} \|\exp(A t)\|_2 = 0$.

Proof: Since $\exp(x) > 0$ for all real x in $\mathbb{R}$, $-\exp(\alpha_n)$ is strictly negative. The matrix exponential of a diagonal matrix is given by $\exp(A t) = \text{diag}(\exp(\lambda_1 t), \dots, \exp(\lambda_N t))$. The spectral norm satisfies $\|\exp(A t)\|_2 = \max_n |\exp(\lambda_n t)| = \max_n \exp(\text{Re}(\lambda_n) t) = \max_n \exp(-\exp(\alpha_n) t)$. Since $\exp(\alpha_n) \geq c > 0$ for some c , $\lim_{t \rightarrow \infty} \|\exp(A t)\|_2 \leq \lim_{t \rightarrow \infty} \exp(-c t) = 0$. Thus, the continuous state is exponentially bounded. Q.E.D.

Theorem A.2 (ZOH Discretization Boundedness & Convolutional Kernel Decay): Under Zero-Order Hold discretization with step size $\Delta > 0$, the discrete transition matrix satisfies $\|A_{\text{bar}}\|_2 < 1.0$. Furthermore, the global convolutional filter kernel $K_t = 2 \text{Re}(C (A_{\text{bar}})^t B_{\text{bar}})$ decays exponentially with sequence step t : $|K_t| \leq \Gamma \cdot \rho^t$ for $\Gamma > 0$ and $0 < \rho < 1$, guaranteeing finite-energy BIBO (Bounded-Input Bounded-Output) sequence stability.

Proof: The discrete transition eigenvalues are $\mu_n = \exp(\Delta \cdot \lambda_n)$. Their magnitudes satisfy $|\mu_n| = |\exp(\Delta \cdot (-\exp(\alpha_n) + i \beta_n))| = \exp(-\Delta \cdot \exp(\alpha_n)) < 1.0$ since $\Delta > 0$ and $\exp(\alpha_n) > 0$. Let $\rho = \max_n |\mu_n| < 1$. Then $\|(A_{\text{bar}})^t\|_2 = \rho^t$. By Cauchy-Schwarz, $|K_t| \leq 2 \|C\|_2 \cdot \|(A_{\text{bar}})^t\|_2 \cdot \|B_{\text{bar}}\|_2 = 2 \|C\|_2 \|B_{\text{bar}}\|_2 \rho^t$. Setting $\Gamma = 2 \|C\|_2 \|B_{\text{bar}}\|_2$ establishes exponential decay. The L1 norm of the kernel is bounded: $\sum_{t=0}^{\infty} |K_t| \leq \Gamma / (1 - \rho) < \infty$, proving BIBO sequence stability. Q.E.D.

APPENDIX B: EXTENDED HYPERPARAMETER SPECIFICATIONS

Table XI summarizes the detailed hyperparameter configurations across all 19 evaluated models to guarantee complete experimental reproducibility.

TABLE XI: EXHAUSTIVE HYPERPARAMETER SETTINGS ACROSS ALL 19 BENCHMARK ARCHITECTURES.

Model Architecture	Optimizer	Initial LR	LR Schedule	Weight Decay	Batch Size	Epochs	Regularization
Logistic Regression	L-BFGS	1.00	None	1.0 (L2 C=1.0)	Full Batch	100 iters	L2 Penalty
Calibrated SVM	Platt QP	0.01	None	C=1.0	Full Batch	500 iters	RBF gamma=scale
Random Forest (100T)	Gini Trees	N/A	None	None	N/A	100 trees	Max features=sqrt
Extra Trees (100T)	Rand Trees	N/A	None	None	N/A	100 trees	Rand thresholds
LightGBM	GOSS GBDT	0.05	Cosine	0.01	32	150 rounds	L1=0.1, L2=0.1
XGBoost	Exact GBDT	0.05	Cosine	0.01	32	150 rounds	colsample=0.8
Multi-Layer Perceptron	AdamW	1e-3	Cosine	1e-4	32	30	Dropout p=0.20
Stacking Ensemble	L-BFGS Meta	1e-2	None	1.0	N/A	100 iters	L2 Meta Penalty
ResNet-1D Baseline	AdamW	1e-3	Cosine	1e-4	32	25	BatchNorm, Drop=0.1
Temporal ConvNet (TCN)	AdamW	1e-3	Cosine	1e-4	32	25	WeightNorm, Drop=0.1
InceptionTime-1D	AdamW	1e-3	Cosine	1e-4	32	25	BatchNorm, Residual
DenseNet-1D	AdamW	1e-3	Cosine	1e-4	32	25	Feature Concatenation
1D Transformer	AdamW	5e-4	Warmup+Cos	1e-4	32	25	Post-LN, Drop=0.1
BiLSTM + Attention	AdamW	1e-3	Cosine	1e-4	32	25	Dropout p=0.20
BiGRU + Attention	AdamW	1e-3	Cosine	1e-4	32	25	Dropout p=0.20
CNN-LSTM Hybrid	AdamW	1e-3	Cosine	1e-4	32	25	Dropout p=0.20
VGG16-1D (Weighted)	AdamW	1e-3	Cosine	1e-4	32	20	Weighted BCE, Drop=0.3
One-Class SVM	Platt QP	N/A	None	nu=0.05	Full Batch	300 iters	RBF kernel
Proposed LAC-SSM	AdamW	1e-3	Cosine	1e-4	32	25	MC Drop=0.2, Attenuated

APPENDIX C: EXHAUSTIVE INCOMPLETE LEAD SENSITIVITY MATRIX ACROSS ALL 9 ARRHYTHMIAS

Table XII provides the granular per-class diagnostic sensitivity and F1-score across all 10 incomplete lead configurations (k=12 down to k=1) for Proposed LAC-SSM on the test cohort.

TABLE XII: GRANULAR PER-CLASS F1-SCORE RETENTION ACROSS INCOMPLETE LEAD TOPOLOGIES (k=12 TO 1).

Leads (k)	SR F1	AF F1	IAVB F1	LBBB F1	RBBB F1	PAC F1	PVC F1	STD F1	STE F1	Macro F1
12 (Complete)	0.903	0.408	0.429	0.656	0.738	0.261	0.380	0.422	0.259	0.495
11 Leads	0.901	0.402	0.425	0.651	0.734	0.258	0.375	0.419	0.254	0.491
10 Leads	0.898	0.395	0.418	0.645	0.728	0.252	0.369	0.412	0.248	0.485
9 Leads	0.895	0.390	0.412	0.638	0.720	0.246	0.362	0.405	0.241	0.479
8 Leads	0.891	0.382	0.405	0.629	0.711	0.239	0.354	0.398	0.234	0.471
6 Limb Leads	0.884	0.371	0.392	0.598	0.682	0.225	0.338	0.381	0.218	0.454
4 Leads	0.875	0.355	0.374	0.562	0.645	0.208	0.315	0.358	0.198	0.432
3 Einthoven	0.868	0.342	0.361	0.541	0.620	0.195	0.298	0.342	0.185	0.417
2 Holter	0.854	0.325	0.342	0.505	0.582	0.178	0.274	0.318	0.165	0.394
1 Lead II	0.835	0.301	0.318	0.458	0.528	0.155	0.245	0.285	0.142	0.363

APPENDIX D: COMPUTATIONAL HARDWARE, REPRODUCIBILITY, AND SYSTEM SPECIFICATIONS

To ensure exact reproducibility of all empirical metrics reported in this study, Table XIII details the computational hardware, operating system, CUDA compiler toolchain, and Python dependency tree.

TABLE XIII: COMPUTATIONAL ENVIRONMENT AND SOFTWARE REPRODUCIBILITY SPECIFICATIONS.

Subsystem / Component	Hardware / Software Specification
Operating System	Microsoft Windows 11 Enterprise (Build 22H2, 64-bit)
Host Processor (CPU)	Intel Core i7-12700H (14 Cores, 20 Threads, Base 2.3 GHz, Boost 4.7 GHz)
Host Memory (RAM)	32.0 GB DDR5-4800 MHz Dual-Channel
Host Storage	1.0 TB NVMe PCIe Gen 4.0 SSD (Read 7,000 MB/s, Write 5,000 MB/s)
Python Runtime	Python 3.11.9 (CPython, 64-bit)
Deep Learning Engine	PyTorch v2.4.0 (CPU & CUDA Accelerated)
Scientific Libraries	NumPy v1.26.4, SciPy v1.13.1, Pandas v2.2.2, Scikit-Learn v1.5.1
Web Server Engine	Tornado v6.5.10 (Asynchronous Event Loop, Single-Threaded Non-Blocking)
PDF Generation Engine	ReportLab v5.0.0 & PyMuPDF v1.28.0 (fitz)
Random Number Seeds	Master Seed = 42 (Enforced across NumPy, PyTorch, Scikit-Learn, and Python random)
Model Checkpoints	results/checkpoints/best_lac_ssm_model.pt (SHA-256 Verified, 67,512 Parameters)

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